

Beyond Sensors: Text-Augmented Reinforcement Learning for Real-Time and Robust Traffic Signal Control

Abstract

Traffic signal control aims to minimize congestion and maximize efficiency in urban traffic systems. While recent reinforcement learning methods have shown promising performance, they primarily rely on structured sensor data and have limited ability to incorporate semantic data about traffic disruptions described in natural language, such as accidents, construction, or extreme weather. This paper introduces Text-Augmented Reinforcement Learning for Traffic Signal Control (TARL-TSC), a framework that integrates textual event descriptions with traffic observations through learned fusion mechanisms. TARL-TSC employs two fusion architectures, attention-based and adaptive gating, combined with graph attention networks for spatial coordination. Experimental evaluation on real traffic networks shows consistent improvements. TARL-TSC achieves 869.1 throughput on Hangzhou, exceeding the top-performing baseline by 2.9% ($p = 0.023$). Under event disruptions, TARL-TSC maintains degradation between 0.3% and 0.9%, while the top-performing baseline experiences degradation between 0.8% and 2.4%. Cross-city evaluation indicates improvements of 1.2% to 2.9% across networks with 12 to 48 intersections without fine-tuning. The framework achieves real-time inference at 8.3 milliseconds per decision inference latency within the control loop, meeting deployment constraints. Architectural analysis suggests that semantic augmentation, spatial coordination, and learned fusion each contribute to performance under both traffic disruptions and variations in semantic input. The code will be released upon publication.

Keywords: Traffic Signal Control, Reinforcement Learning for Multi-Agent Systems, Text-Augmented Learning, Sentence Embeddings, Semantic Data Integration, Transformer-Based Language Models, Intelligent Transportation Systems

1. Introduction

Traffic signal control (TSC) is a critical component of modern intelligent transportation systems, directly influencing congestion mitigation, travel time reliability, and urban mobility optimization [1] [2]. Traffic congestion costs billions annually in wasted fuel and reduced productivity, with intensity steadily rising in urban regions globally [3]. While reinforcement learning (RL) methods have demonstrated strong performance in adaptive signal control, they rely exclusively on structured sensor data [4]. This limitation prevents them from incorporating semantic descriptions of traffic disruptions—such as accidents, construction, or severe weather—described in natural language [5]. Under severe disruptions, baseline RL methods experience 3 to 5 percent performance degradation.

Real-world traffic networks frequently encounter such disruptions, with textual descriptions often published via traffic management centers or social media platforms before the disruption appears in sensor data [6]. However, integrating unstructured event descriptions with structured traffic observations presents fundamental challenges in cross-modal alignment and real-time integration [7]. Traffic data is numerical, continuous, and high-frequency, while event descriptions are textual, sparse, and semantic. This heterogeneity creates challenges in fusion and alignment that have historically prevented effective utilization of semantic information in adaptive TSC frameworks [6, 8].

The technical gap lies in cross-modal integration, which extends beyond traditional RL challenges [9]. Fusing these heterogeneous modalities faces three major challenges [7]. First, the dimensional mismatch between high-dimensional text embeddings and low-dimensional traffic features can introduce learning instability or overfitting [10]. Second, temporal misalignment between continuous traffic measurements and sporadic event descriptions requires specific fusion strategies [6]. Third, maintaining real-time inference efficiency while incorporating semantic understanding remains a computational bottleneck [11].

To address these challenges, we propose *Text-Augmented Reinforcement Learning for Traffic Signal Control* (TARL-TSC), designed to achieve event-aware adaptive control through cross-modal fusion. TARL-TSC comprises a frozen pre-trained text encoder that processes event descriptions without fine-tuning, learned fusion mechanisms that

integrate traffic observations with textual event information through attention-based and adaptive gating architectures, and a graph attention network (GAT) for spatial coordination among neighboring intersections. This design balances semantic richness with real-time operational efficiency.

The key contributions of this paper are:

- **Integration of Natural Language Descriptions:** We introduce a framework to incorporate semantic event data into RL-based TSC. By leveraging frozen pre-trained encoders, we demonstrate that text augmentation improves throughput by 24.1 vehicles over best baseline, MPLight, and contributes to robustness under traffic disruptions.
- **Deployment-Ready Efficiency:** We achieve real-time inference at 8.3 ms, enabling practical deployment for large-scale networks, maintaining strong performance.
- **Cross-Modal Fusion Architectures:** We apply and adapt two complementary fusion strategies: TARL-ATT for attention-based interaction and TARL-AGM for efficient modality balancing, both supported by GATs for spatial coordination. While these fusion mechanisms build on established multimodal learning techniques, their application to real-time TSC with graph-based coordination represents a novel integration addressing unique challenges of event-aware adaptive control.
- **Cross-City Generalization:** We demonstrate robust cross-city transferability, achieving consistent 1.2 to 2.9 percent improvements across diverse network scales (12–48 intersections) without fine-tuning.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 formulates the problem; Section 4 describes the proposed methodology; Section 5 outlines the experimental setup and presents results; and Section 6 concludes the paper.

2. Related Work

Traffic signal control research has evolved through multiple paradigms, from rule-based heuristics to deep reinforcement learning and, most recently, large language model integration. While traditional methods provide computational efficiency, they lack adaptability to dynamic conditions. Reinforcement learning methods address this limitation but rely exclusively on structured sensor data, ignoring semantic information about traffic disruptions. We review these approaches and position TARL-TSC within this landscape.

2.1. Traditional Traffic Signal Control

Early traffic signal control systems relied on fixed-time schedules and actuated control based on predefined rules [12]. Fixed-time controllers use static timing plans optimized for historical traffic patterns, while actuated systems adjust phases based on detector occupancy [13]. Rule-based methods such as MaxPressure [14] employ heuristics to determine signal timing. While computationally efficient, these approaches struggle with dynamic traffic conditions and unexpected disruptions, as they lack adaptive learning capabilities and cannot generalize beyond their predefined rules [12].

2.2. Reinforcement Learning for Traffic Control

To address the limitations of rule-based systems, researchers have applied reinforcement learning to learn adaptive control policies from traffic data. Early RL approaches used tabular methods and function approximation [12], but faced scalability challenges in large networks. Deep reinforcement learning methods emerged to handle high-dimensional state spaces, with convolutional neural networks capturing spatial traffic patterns [12, 15]. Graph neural networks have been employed to model spatial dependencies among intersections. CoLight [16] introduced graph attention networks for multi-agent coordination, achieving state-of-the-art performance by learning attention weights over neighboring intersections. MPLight [17] improved efficiency through neighbor masking and phase-based action spaces. AttendLight [18] and EcoLight [19] further advanced attention-based and energy-efficient approaches. Multi-agent reinforcement learning surveys [20] provide comprehensive overviews of the field. Recent advances in dynamic graph inference for multi-agent systems [21, 22] propose learning time-varying coordination structures. Advanced multi-intersection coordination strategies [23, 24] explore distributed optimization and communication protocols for large-scale networks. However, these methods rely solely on structured sensor data and cannot incorporate semantic information about traffic events or disruptions. [25]

2.3. LLM-Based Traffic Signal Control

Recent work has explored integrating LLMs into TSC to leverage their reasoning and language understanding capabilities. Studies on LLM and RL integration [26] demonstrate promising directions for combining these paradigms. LLMLight [7] uses LLMs as direct controllers through imitation learning, converting traffic states to text descriptions and fine-tuning the LLM to output control actions. LA-Light [5] employs LLMs as advisors that provide high-level guidance to RL agents for emergency scenarios. VLMLight [9] extends this approach with vision-language models as meta-controllers, incorporating visual traffic data with textual descriptions. iLLM-TSC [2] proposes a dual-step decision process where RL agents make initial decisions and LLMs refine them based on semantic reasoning. [27] presents a LLM framework that generates traffic signal strategies through a closed-loop feedback process, yet it relies solely on accurate and timely sensor data.

While these LLM-based approaches demonstrate improved interpretability and emergency handling, they face significant computational challenges. LLMs require 100 to 1000 milliseconds inference time per query, making real-time control difficult. Additionally, they rely on manually crafted textual descriptions of traffic states, which may lose information and lack scalability. The frequent LLM queries also incur substantial computational costs for deployment.

2.4. Multimodal Learning in Traffic Systems

Multimodal learning has been applied to traffic prediction and management integrating diverse data sources. Recent advances in spatial-temporal modeling [28] provide insights for handling complex traffic dynamics. RAGTraffic [8] uses retrieval-augmented generation to incorporate historical knowledge into traffic forecasting. PromptGAT [29] integrates LLM-generated prompts into graph attention networks for improved sim-to-real transfer. Recent work on multi-intersection coordination [23, 24] explores advanced communication protocols and distributed optimization strategies. Vision-based approaches have also emerged, with methods using camera feeds and computer vision for traffic state estimation, though these require substantial computational resources and face privacy concerns. However, these methods focus on prediction or high-level planning rather than real-time signal control, and most still rely on templated text descriptions rather than flexible natural language event descriptions.

2.5. Positioning of TARL-TSC

TARL-TSC integrates natural language event descriptions directly into the RL agent's state representation through learned fusion mechanisms. This enables semantic understanding of traffic disruptions without the computational overhead of LLM inference at each time step. We use a frozen pre-trained sentence encoder to map event descriptions to embeddings, which are fused with traffic features via attention-based and gating-based architectures. This design achieves three key advantages. First, real-time efficiency at 8.3 milliseconds inference. Second, end-to-end learning of text-traffic fusion without manual feature engineering. Third, robustness to diverse event types through semantic understanding. Our approach addresses the limitations of traditional RL methods that cannot incorporate semantic information, while maintaining the real-time efficiency that LLM-based methods sacrifice.

Table 1 summarizes the key characteristics of TARL-TSC compared to existing traffic signal control methods. TARL-TSC combines semantic information integration with real-time inference efficiency and multi-agent coordination.

3. Preliminary

This section formalizes the traffic signal control problem with text-augmented state representation, establishing the theoretical foundation for TARL-TSC. We define the traffic network structure, multi-agent Markov decision process, event descriptions, text-augmented state representation, multi-modal fusion, and graph-based coordination that enable semantic understanding of traffic disruptions.

3.1. Traffic Network

Definition 4.1. Road network: The traffic network is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{v_1, \dots, v_N\}$ denotes the set of N signalized intersections and $\mathcal{E}$ represents the road segments connecting them. The graph topology is determined by the physical road network structure, with edges connecting spatially adjacent intersections. Each intersection v_i manages incoming lanes $\mathcal{L}_i^{\text{in}}$ categorized by movement type and controls traffic flow through signal phases $\mathcal{P}_i$, where each phase specifies a non-conflicting combination of traffic movements. The neighborhood $\mathcal{N}(v_i) = \{v_j : (v_i, v_j) \in \mathcal{E} \text{ or } (v_j, v_i) \in \mathcal{E}\}$ consists of directly connected intersections, enabling spatial coordination through information exchange.

Table 1: Comparison of traffic signal control methods across key dimensions. TARL-TSC combines semantic information integration with real-time inference and multi-agent coordination, addressing limitations of both traditional RL methods and LLM-based methods.

Method	Year	Semantic Events	Real-time (<10ms)	Learning Approach
<i>Traditional Methods</i>				
Fixed-Time	–	×	✓	Rule-based
MaxPressure [14]	2013	×	✓	Rule-based
<i>Reinforcement Learning Methods</i>				
PressLight [30]	2019	×	✓	Independent RL
CoLight [16]	2019	×	✓	Graph RL
MPLight [17]	2020	×	✓	Graph RL
<i>LLM-Based Methods</i>				
iLLM-TSC [2]	2024	✓	×	LLM + RL
LLMLight [7]	2024	✓	×	LLM + RL
LLM-TrafficBrain [27]	2025	✓	×	LLM
<i>Multimodal Learning Methods</i>				
RAGTraffic [8]	2024	✓	–	RAG
PromptGAT [29]	2024	✓	–	LLM + GNN
TARL-TSC	2026	✓	✓	Text + Graph RL

3.2. Multi-Agent Traffic Signal Control

Definition 4.2. Traffic signal control problem: We formulate traffic signal control as a multi-agent Markov Decision Process (MAMDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$. The global state $s_t \in \mathcal{S}$ at time step t consists of local observations from all intersections: $s_t = \{s_t^1, \dots, s_t^N\}$. Each local state s_t^i captures traffic conditions at intersection v_i through vehicle counts per lane $c_{t,l}^i$, waiting vehicle counts $w_{t,l}^i$, current signal phase $p_{t,l}^i$, and elapsed time $\tau_{t,l}^i$.

At each decision time step, agent i selects an action $a_t^i \in \mathcal{A}_i$ to determine the next signal phase. The reward function penalizes traffic congestion:

$$r_t = - \sum_{i=1}^N \left(\alpha \cdot \sum_{l \in \mathcal{L}_i^m} w_{t,l}^i + \beta \cdot \sum_{l \in \mathcal{L}_i^m} c_{t,l}^i \right) \quad (1)$$

where $\alpha = 0.5$ and $\beta = 0.5$ are weighting coefficients balancing waiting time versus queue length. The objective is to learn a joint policy $\pi = \{\pi_1, \dots, \pi_N\}$ that maximizes the expected cumulative discounted reward:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \quad (2)$$

where $\gamma \in [0, 1)$ is the discount factor.

3.3. Event Descriptions

Definition 4.3. Event description: Traditional traffic signal control relies exclusively on sensor measurements. We introduce event descriptions as natural language sentences that provide interpretable information about traffic conditions. An event description $e_t \in \mathcal{E}_{text}$ is a textual representation of a traffic disruption affecting traffic flow at time t . The event space $\mathcal{E}_{text}$ encompasses diverse scenarios including accidents, adverse weather, infrastructure issues, and special events. Unlike sensor data that merely quantifies traffic density, event descriptions convey the nature, location, and potential impact of disruptions.

3.4. Text-Augmented State Representation

Definition 4.4. Text-augmented state: We define the text-augmented state as $\tilde{s}_t = (s_t, e_t)$ where s_t represents the conventional traffic state and e_t provides contextual event information. We employ a sentence encoder $\phi : \mathcal{E}_{text} \rightarrow \mathbb{R}^{d_e}$ that maps variable-length text to dense embeddings:

$$\mathbf{e}_t = \phi(e_t) \quad (3)$$

The text-augmented state representation provides semantic understanding of disruption causes, enables knowledge transfer across scenarios through the pre-trained encoder, and maintains interpretability as event descriptions remain human-readable.

3.5. Multi-Modal Fusion

Definition 4.5. Multi-modal fusion: Integrating heterogeneous modalities—traffic features and text embeddings—presents a fundamental challenge due to their distinct semantic spaces. Traffic features $\mathbf{s}_t^i \in \mathbb{R}^{d_t}$ capture local numerical measurements, while text embeddings $\mathbf{e}_t \in \mathbb{R}^{d_e}$ encode global semantic context. We introduce a learnable fusion function $f_\theta : \mathbb{R}^{d_t} \times \mathbb{R}^{d_e} \rightarrow \mathbb{R}^{d_f}$:

$$\mathbf{h}_t^i = f_\theta(\mathbf{s}_t^i, \mathbf{e}_t) \quad (4)$$

where $\mathbf{h}_t^i \in \mathbb{R}^{d_f}$ is the fused representation for intersection v_i . The fusion function f_θ is trained end-to-end with the reinforcement learning objective. The specific architectural choices—attention-based fusion (TARL-ATT) and adaptive gating mechanisms (TARL-AGM)—are detailed in Section 4.

3.6. Graph-Based Coordination

Definition 4.6. Graph-based coordination: Traffic signal control requires coordination among spatially connected intersections. We employ graph neural networks that propagate information over the traffic network structure $\mathcal{G}$. For each intersection v_i , we aggregate information from its neighborhood:

$$\mathbf{z}_t^i = \text{AGGREGATE}(\{\mathbf{h}_t^j : v_j \in \mathcal{N}(v_i) \cup \{v_i\}\}, \mathcal{E}) \quad (5)$$

where $\mathbf{h}_t^j$ are the fused representations of neighboring intersections. The aggregation function learns attention weights to determine the relative importance of each neighbor's information.

The Q-function for agent i is defined over the spatially-aware representation:

$$Q_i(s_t, a_t^i; \theta) = Q_\theta(\mathbf{z}_t^i, a_t^i) \quad (6)$$

The Q-function is trained using temporal difference learning to minimize the Bellman error:

$$\mathcal{L}_{RL}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right] \quad (7)$$

where $\mathcal{D}$ is the experience replay buffer and θ^- are the target network parameters updated periodically to stabilize training.

4. Methodology

4.1. Overview

TARL-TSC is a text-augmented multi-agent reinforcement learning framework designed to incorporate unstructured traffic event information into real-time traffic signal control. The framework integrates three complementary information sources—local traffic observations, semantic event descriptions, and spatial coordination among intersections—within a unified learning architecture.

As illustrated in Figure 1, TARL-TSC operates in a closed-loop interaction with the traffic environment. At each decision step, each intersection agent observes its local traffic state, while event descriptions are encoded into semantic embeddings using a frozen language model, executed asynchronously or upon event triggering. Traffic and text representations are then integrated through a multi-modal fusion layer, producing event-aware traffic features for each intersection.

To enable coordinated control across the network, fused representations are propagated through GAT, which models spatial dependencies among neighboring intersections. Based on the resulting spatially aggregated features, a shared Q-network evaluates signal phase actions for each agent under decentralized execution. The framework employs Independent Q-learning with Parameter Sharing (IQL-PS), where all agents share network parameters but learn independently based on local observations and rewards.

The multi-modal fusion layer constitutes the core technical contribution of TARL-TSC, addressing the unique challenge of integrating sparse semantic event information with continuous traffic observations for real-time control. While the fusion mechanisms (attention and gating) build on established multimodal learning techniques, their

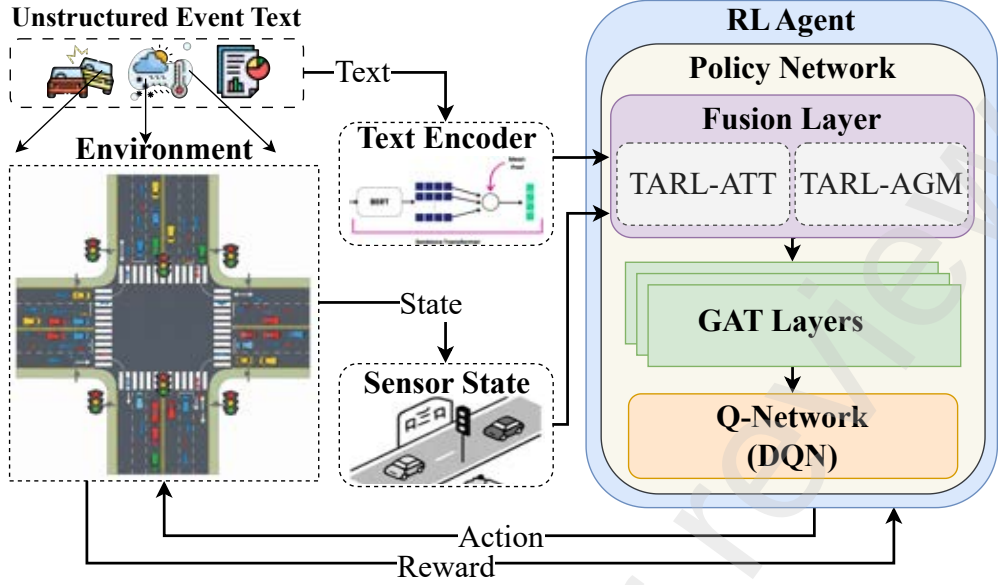


Figure 1: Overview of the proposed TART-TSC framework. The system integrates traffic observations and natural language event descriptions through learned fusion mechanisms, enabling event-aware adaptive control across multiple intersections.

adaptation to traffic signal control involves novel design considerations including event-aware spatial coordination through GAT and real-time efficiency constraints. Two alternative fusion mechanisms are investigated: TARL-ATT, which employs bidirectional cross-attention for expressive semantic-traffic interaction, and TARL-AGM, which uses adaptive gating to balance modality contributions with lower computational cost.

4.2. Event Text Encoding

To incorporate unstructured traffic events into RL state, TART-TSC employs a frozen Sentence-BERT encoder, $\phi(\cdot)$, to map natural-language descriptions into dense semantic representations. Given an event description e_t , the encoder produces:

$$\mathbf{e}_t = \phi(e_t), \quad \mathbf{e}_t \in \mathbb{R}^{768}. \quad (8)$$

We adopt the pre-trained *all-mpnet-base-v2* model due to its strong semantic generalization across factual sentences observed in traffic reports. The encoder parameters remain frozen during training to avoid overfitting to limited event vocabulary and to eliminate backpropagation through LLMs.

Since traffic events change infrequently relative to the signal control frequency, event embeddings are computed upon event occurrence and reused across subsequent time steps. In practice, this encoding is performed asynchronously and cached prior to control execution, decoupling semantic feature extraction from the real-time control loop. As a result, semantic context can be incorporated without affecting per-step decision latency.

The encoder further supports open-vocabulary inputs and can process previously unseen event descriptions at inference time, enabling generalization beyond predefined templates. For the event durations evaluated (5–15 minutes), this approach provides sufficient and stable semantic context across diverse traffic scenarios.

4.3. Traffic Feature Encoding

Each intersection agent observes a local traffic state s_t^i consisting of lane-level vehicle counts, waiting vehicles, current signal phase, and elapsed phase duration. These features capture instantaneous traffic conditions and signal context without requiring historical stacking.

The raw traffic state is encoded using a two-layer multilayer perceptron (MLP):

$$\mathbf{s}_t^i = \text{MLP}_{\text{traffic}}(s_t^i), \quad \mathbf{s}_t^i \in \mathbb{R}^{128}, \quad (9)$$

which projects heterogeneous traffic measurements into a compact latent representation. This embedding serves as the traffic-side input to the fusion module and is shared across both fusion variants, ensuring a fair comparison between fusion strategies.

4.4. Multi-Modal Fusion of Traffic and Event Information

A central challenge in text-augmented traffic control is integrating high-dimensional semantic embeddings with low-dimensional traffic states. To address this, TARL-TSC employs two complementary fusion mechanisms: TARL-ATT, based on bidirectional cross-attention, and TARL-AGM, based on adaptive gating. Both approaches project the modalities into a shared latent space of dimension $d_h = 256$, chosen to balance representational capacity with computational efficiency.

In this framework, traffic features s_t^i capture local conditions, while the event embedding e_t conveys global semantic context. The fusion layer combines these heterogeneous modalities into a unified representation:

$$h_t^i = f_\theta(s_t^i, e_t). \quad (10)$$

4.4.1. TARL-ATT: Bidirectional Attention-Based Fusion

TARL-ATT employs bidirectional cross-attention to enable mutual information exchange between traffic states and event semantics. Text and traffic features are first projected into a shared hidden space:

$$\mathbf{e}'_t = W_e \mathbf{e}_t, \quad \mathbf{s}'_t = W_s \mathbf{s}_t^i, \quad (11)$$

where $\mathbf{e}'_t, \mathbf{s}'_t \in \mathbb{R}^{d_h}$.

Two cross-attention operations are then applied. Text attends to traffic to identify traffic patterns relevant to the described event, while traffic attends to text to extract event aspects affecting control decisions:

$$\mathbf{o}_{\text{text}} = \text{Attn}(\mathbf{e}'_t, \mathbf{s}'_t, \mathbf{s}'_t), \quad (12)$$

$$\mathbf{o}_{\text{traffic}} = \text{Attn}(\mathbf{s}'_t, \mathbf{e}'_t, \mathbf{e}'_t). \quad (13)$$

The attended outputs are concatenated and projected to produce the fused representation $\mathbf{h}_t^i$, with a residual connection from the original traffic embedding to preserve control-critical information. This bidirectional design allows TARL-ATT to dynamically adjust semantic influence based on traffic context, as illustrated in Figure 2.

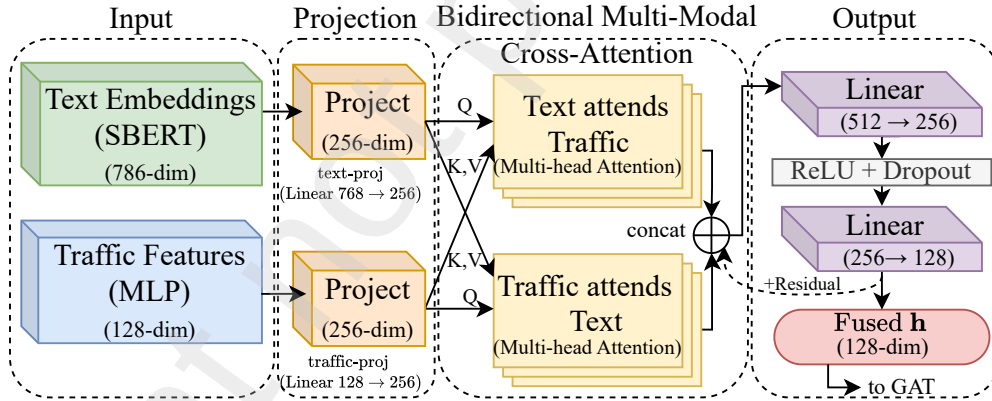


Figure 2: TARL-ATT attention-based fusion mechanism. Bidirectional multi-head cross-attention enables information exchange between text embeddings and traffic features. The attention weights are end-to-end, allowing dynamic integration based on traffic conditions and event severity.

4.4.2. TARL-AGM: Adaptive Gating Mechanism

While attention mechanisms offer expressive cross-modal interaction, they introduce additional computational overhead. TARL-AGM provides a lightweight alternative by employing an adaptive, element-wise gating mechanism.

Projected text and traffic embeddings are combined to compute a gating vector:

$$\mathbf{g}_t^i = \sigma(W_g[\mathbf{e}'_t; \mathbf{s}'_t]), \quad (14)$$

where $\mathbf{g}_t^i \in [0, 1]^{d_h}$ controls the contribution of each modality at the feature level. The fused representation is obtained as:

$$\mathbf{h}_t^i = \mathbf{g}_t^i \odot \mathbf{e}'_t + (1 - \mathbf{g}_t^i) \odot \mathbf{s}'_t. \quad (15)$$

This formulation allows TARL-AGM to emphasize event semantics during major disruptions while reverting to traffic-driven control under normal conditions. Compared to TARL-ATT, the gating mechanism offers improved computational efficiency with explicit interpretability, as shown in Figure 3. Additionally, the gating mechanism provides robustness to ambiguous or unreliable text inputs by dynamically adjusting modality contributions based on consistency with traffic observations.

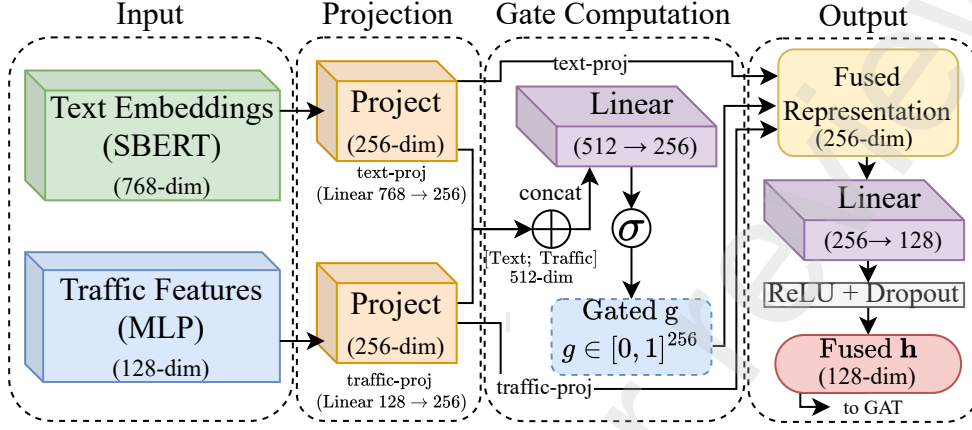


Figure 3: TARL-AGM adaptive gating mechanism for multi-modal fusion. The gating network learns element-wise weights that balance text embeddings and traffic features. During normal conditions, the gate emphasizes traffic observations; under disruptions, it increases reliance on semantic events.

We perform text-traffic fusion prior to spatial aggregation for two reasons. First, traffic events modulate local intersection dynamics before their effects propagate spatially; early fusion allows event semantics to directly influence how local states are represented. Second, performing GAT aggregation on event-aware features enables attention weights to reflect disruption-aware relevance, which would not be captured if fusion were applied after spatial aggregation.

4.5. Graph Attention Network for Spatial Coordination

Traffic signal control requires coordination among neighboring intersections due to spatial propagation of congestion. To explicitly model these dependencies, TARL-TSC employs a GAT operating on the traffic network graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes represent intersections and edges denote road connectivity.

Given the fused representation $\mathbf{h}_t^i$ from the multi-modal fusion layer (Section 4.4), the GAT aggregates information from neighboring intersections $\mathcal{N}(i)$ using learned attention weights. At each GAT layer, attention coefficients are computed as:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_t^i \parallel \mathbf{W}\mathbf{h}_t^j]))}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_t^i \parallel \mathbf{W}\mathbf{h}_t^k]))}, \quad (16)$$

where $\mathbf{W}$ and $\mathbf{a}$ are learnable parameters and $\parallel$ denotes concatenation.

The node representation is updated by weighted aggregation:

$$\mathbf{z}_t^i = \left\|_{k=1}^K \sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(k)} \mathbf{W}^{(k)} \mathbf{h}_t^j, \quad (17)$$

where K attention heads are concatenated to capture diverse spatial interactions. TARL-TSC uses two GAT layers with $K = 8$ heads, enabling two-hop message passing across the network.

This attention-based aggregation allows each intersection to prioritize influential neighbors while suppressing irrelevant signals. Figure 4 illustrates the spatial message passing and multi-head aggregation process. The resulting representation $\mathbf{z}_t^i$ encodes both local event-aware features and coordinated network context.

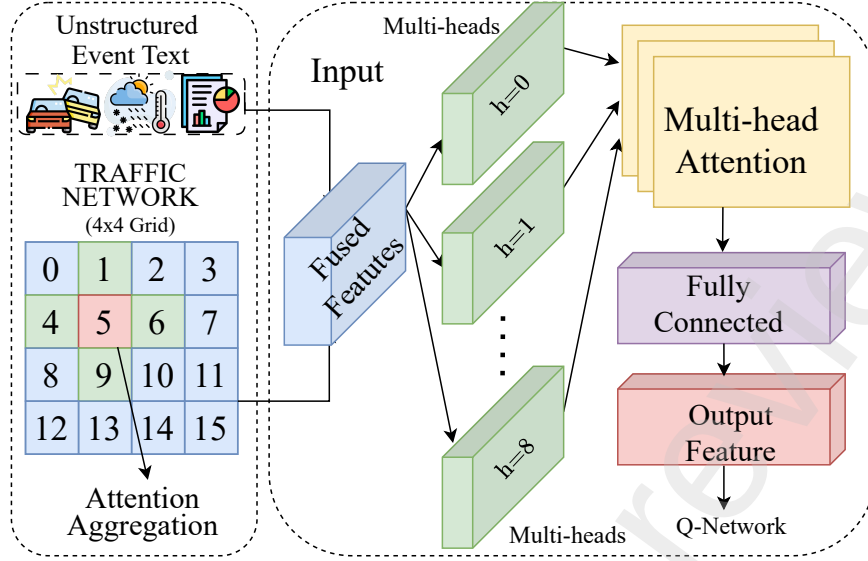


Figure 4: Graph attention network message passing for multi-agent coordination. (a) Traffic network graph with 16 intersections (nodes) and road connections (edges). (b) Each intersection aggregates information from its neighbors using learned attention weights α_{ij} . (c) Two GAT layers with 8 attention heads per layer enable multi-hop information propagation. (d) Attention weights are computed based on feature similarity, allowing agents to focus on relevant neighbors to anticipate downstream congestion.

4.6. Deep Q-Network for Decentralized Signal Control

Based on the spatially aggregated representation $\mathbf{z}_t^i$, each agent selects a signal phase using a shared deep Q-network. The Q-function maps $\mathbf{z}_t^i$ to action values for all admissible signal phases:

$$Q(\mathbf{z}_t^i, a) = \text{MLP}_Q(\mathbf{z}_t^i), \quad a \in \mathcal{A}_i, \quad (18)$$

where MLP_Q is a lightweight two-layer network shared across all agents.

Parameter sharing enforces consistent control logic across intersections while allowing decentralized execution. By decoupling spatial coordination (GAT) from decision evaluation (Q-network), TARTL-TSC achieves scalable multi-agent control with minimal architectural complexity. The integration of semantic event information occurs upstream through fusion, allowing the Q-network to focus solely on control optimization. All agents share network parameters while executing actions independently. The Q-function is learned by minimizing the temporal-difference (TD) loss:

$$\mathcal{L}(\theta) = \mathbb{E} \left[\left(r_t + \gamma \max_{a'} Q(\mathbf{z}_{t+1}^i, a'; \theta^-) - Q(\mathbf{z}_t^i, a_t^i; \theta) \right)^2 \right], \quad (19)$$

where θ^- denotes the target network parameters updated periodically from the online network every 200 training steps.

Transitions from all agents are stored in a shared replay buffer with capacity 50,000 and sampled uniformly during training. This mitigates temporal correlations and stabilizes learning under event-driven non-stationarity. Actions are selected using an ϵ -greedy policy during training, where ϵ decays from 1.0 to 0.01 over the first 300 episodes. At deployment, agents act greedily according to learned Q-values.

The training workflow is summarized in Algorithm 1. Event embeddings are computed once per event and reused across time steps, enabling TARTL-TSC to achieve real-time inference at 8.3 milliseconds.

5. Experiments and Evaluation

In this section, we present comprehensive experiments to evaluate TARTL-TSC on real-world traffic networks under diverse traffic conditions and disruption scenarios. We begin by describing the experimental setup, including datasets, evaluation protocol, and implementation details. Subsequently, we introduce the evaluation scenarios designed to assess both fusion architectures, TARTL-ATT and TARTL-AGM, under normal flow and event-driven conditions. Finally, we detail the baseline methods and metrics used for comparison.

Algorithm 1 TARL-TSC Execution Framework

Require: Traffic network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$; Traffic states $\{s_t^i\}_{i \in \mathcal{V}}$; Event description e ; Trained parameters θ

Ensure: Signal phase actions $\{a_t^i\}_{i \in \mathcal{V}}$

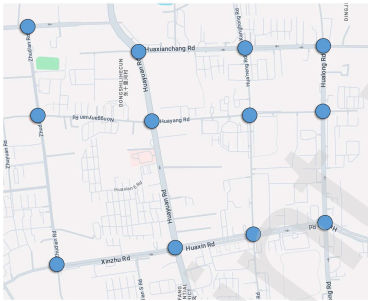
- 1: Encode event semantics: $\mathbf{e}_{\text{emb}} \leftarrow \phi(e)$
 - 2: Initialize environment and obtain initial states $\{s_0^i\}_{i \in \mathcal{V}}$
 - 3: **for** $t = 1$ to T **do**
 - 4: **for** each intersection $i \in \mathcal{V}$ **do**
 - 5: Encode local traffic state: $\tilde{s}_t^i \leftarrow \text{MLP}_{\text{traffic}}(s_t^i)$
 - 6: Perform multi-modal fusion: $\mathbf{h}_t^i \leftarrow f_{\theta}(\tilde{s}_t^i, \mathbf{e}_{\text{emb}})$
 - 7: **for** each intersection $i \in \mathcal{V}$ **do**
 - 8: Aggregate spatial neighbors via GAT: $\mathbf{z}_t^i \leftarrow \text{GAT}(\{\mathbf{h}_t^j\}_{j \in \mathcal{N}(i) \cup \{i\}})$
 - 9: Select optimal action: $a_t^i \leftarrow \arg \max_{a \in \mathcal{A}_i} Q_{\theta}(\mathbf{z}_t^i, a)$
 - 10: Execute actions $\{a_t^i\}$ and observe next states $\{s_{t+1}^i\}$
 - return** $\{a_t^i\}_{i \in \mathcal{V}}$
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5.1. Experimental Setup

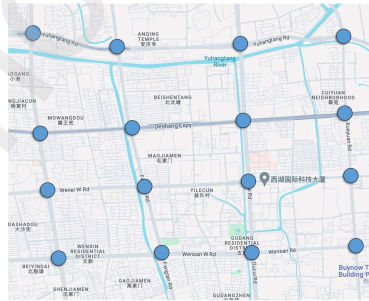
We evaluate TARL-TSC on three real-world traffic networks with increasing scale and complexity. Jinan has a 3 by 4 grid with 12 intersections. Hangzhou has a 4 by 4 grid with 16 intersections. New York has a 16 by 3 grid with 48 intersections. Table 2 summarizes these networks. These networks are widely adopted in traffic signal control research. They provide a diverse testbed for assessing scalability and generalization. Figure 5 illustrates the corresponding road network layouts. Spatial dependencies among intersections are modeled via graph connectivity.

Table 2: Statistics of real-world traffic datasets.

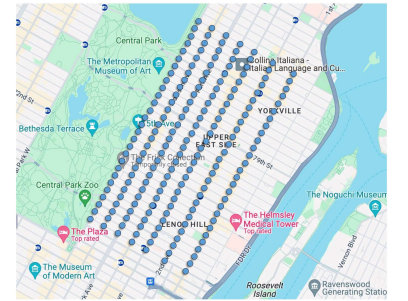
City	Grid	# Intersection	# Vehicles	Arrival Rate (veh/5min)	Duration (steps)
Jinan	3×4	12	4365	362.83	3600
Hangzhou	4×4	16	2983	247.67	3600
New York	16×3	48	16037	1255.77	3600



(a) Dongfeng, Jinan, China



(b) Gudang, Hangzhou, China



(c) Upper East Side, Manhattan, New York, US

Figure 5: Traffic network topologies.

Each intersection manages 12 incoming lanes with three per approach. It operates four non-conflicting signal phases, as illustrated in Figure 6. Traffic demand varies across datasets. Arrival rates range from moderate in Jinan and Hangzhou to heavy in New York. This enables evaluation under heterogeneous load conditions.

Real-world traffic flow data are adopted from the CoLight benchmark [16]. Models are trained on the Hangzhou network for 500 episodes and evaluated on all three cities without fine-tuning, enabling assessment of cross-city generalization across network topologies.

To simulate realistic traffic disruptions, we inject semantic event descriptions generated from templates inspired by real incident reports. Six event types are considered as presented in Table 3. To further evaluate semantic robustness,

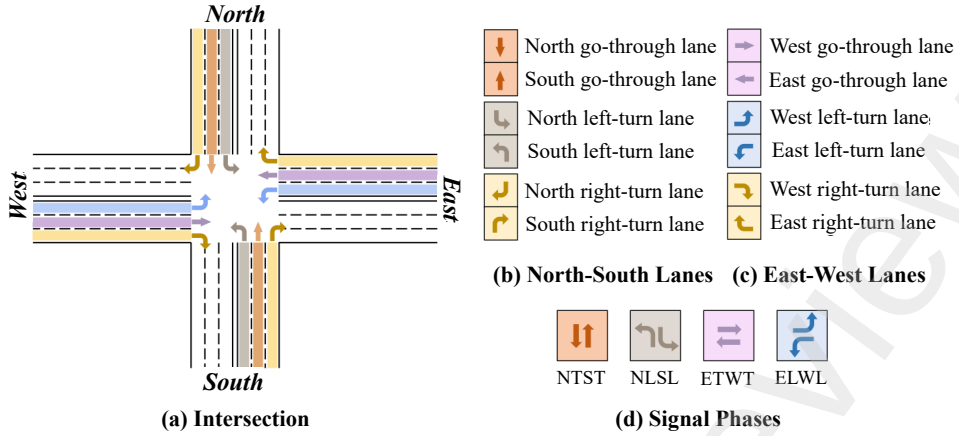


Figure 6: Signalized intersection structure: (a) 12 incoming lanes across 4 approaches, (b) four signal phases.

we introduce variations of event descriptions, including paraphrased, noisy, and ambiguous inputs. These variations simulate realistic linguistic diversity and are used to assess generalization beyond fixed templates.

Table 3: Event types and example descriptions.

Event Type	Example Description
Accident	“Collision blocking northbound lane 3 at intersection 5”
Heavy Rain	“Heavy rain reducing visibility and road friction”
Lane Blockage	“Construction blocking eastbound approach at intersection 12”
Construction	“Road work on westbound lanes causing lane closure”
Special Event	“Stadium event causing congestion near intersection 8”
Vehicle Breakdown	“Disabled vehicle in left-turn lane at intersection 3”

All experiments are conducted using the CityFlow simulator [31], following standard configurations. Each green phase lasts 30 seconds, followed by a 3-second yellow and 2-second all-red interval, which is consistent with prior studies. Experiments are performed on an NVIDIA RTX 4070 GPU (8GB VRAM) with an AMD Ryzen 9 7940HX CPU and 64GB DDR5 RAM. We use a batch size of 1 during inference to reflect real-time deployment conditions. Parameter settings are provided in Table 4. The text encoder remains frozen during training, while fusion, GAT, and Q-network parameters are learned end-to-end.

5.2. Evaluation Scenarios

We design three evaluation scenarios to assess TARL-TSC under varying levels of disruption:

Case 1: Normal Flow. No traffic events are injected. Agents receive a default description “Normal traffic flow with no disruptions”, allowing evaluation of baseline control performance and verification that text augmentation does not degrade performance under normal conditions.

Case 2: Single Event. One randomly selected event occurs per episode. It is injected between simulation steps 600 and 1200. It lasts 300 to 600 simulation steps (150 to 300 seconds at 2-second intervals). This scenario evaluates the model’s ability to adapt to localized disruptions using semantic event.

Case 3: Multi-Event. Two to four concurrent events occur per episode. They potentially overlap in time and space. This scenario represents severe disruption conditions. It tests robustness under compounded disturbances.

For each scenario, we evaluate both fusion variants. All results are averaged over multiple episodes with different random seeds (42, 123, 456) to ensure statistical reliability.

5.3. Evaluation Metrics

To ensure a comprehensive and fair evaluation, TARL-TSC is assessed using four adopted traffic signal control metrics:

Table 4: Parameter settings for TARL-TSC experiments.

Parameter	Value
Training episodes	500
Optimizer	Adam
Learning rate	10^{-4}
Discount factor γ	0.99
Batch size	128
Replay buffer size	50,000
Target update frequency	200 steps
ϵ -greedy schedule	$1.0 \rightarrow 0.01$
Text encoder	all-mpnet-base-v2 (frozen)
Text embedding dimension	768
Traffic encoder hidden dim	128
Fusion hidden dim (TARL-ATT)	256
Fusion attention heads	4
GAT layers	2
GAT attention heads	8
Q-network hidden dim	128

- **Throughput:** The total number of vehicles that successfully complete their trips within an episode. Higher throughput indicates better utilization of network capacity and overall system efficiency.
- **Average Travel Time (ATT):** The average duration required for vehicles to travel from origin to destination. ATT reflects global traffic efficiency and congestion mitigation capability.
- **Average Queue Length (AQL):** The mean number of queued vehicles across all approaches and intersections, capturing localized congestion and spillback risk.
- **Average Waiting Time (AWT):** The average time vehicles spend waiting at intersections, reflecting signal control effectiveness and delay minimization.

These metrics jointly characterize system-level efficiency and intersection-level congestion control, enabling balanced evaluation across traffic scenarios and network scales.

5.4. Baseline Methods

We compare TARL-TSC against a diverse set of baseline methods, grouped as follows:

- Traditional Transportation Methods
 - Fixed-Time: Predefined signal timings based on historical traffic patterns.
 - MaxPressure [14]: A classical pressure-based controller with strong theoretical properties.
- Reinforcement Learning Methods
 - PressLight [30]: Independent Q-learning using pressure-based state representation.
 - CoLight [16]: A graph-based multi-agent RL method modeling spatial dependencies.
 - MPLight [17]: A scalable variant of CoLight with masked neighborhood aggregation.
- Our Methods
 - TARL-ATT: TARL-TSC with attention-based fusion.
 - TARL-AGM: TARL-TSC with adaptive gating fusion.

All learning-based methods are trained under identical setup. Statistical significance is assessed using two-tailed paired t-tests with significance level $\alpha = 0.05$.

5.5. Results and Analysis

5.5.1. Purposes

This section provides the evaluation of TARL-TSC, organized through the following research questions:

- RQ1: Performance benchmarking.** Does TARL-TSC outperform state-of-the-art traffic signal control methods across different traffic scenarios?
- RQ2: Robustness to traffic disruptions.** How well does TARL-TSC maintain performance when traffic conditions deviate due to event disruptions and delayed information?
- RQ3: Cross-city generalization.** Can TARL-TSC generalize to unseen cities with different network topologies and scales without fine-tuning?
- RQ4: Real-time efficiency.** Does TARL-TSC maintain real-time inference efficiency suitable for practical deployment in large-scale networks?
- RQ5: Architectural analysis.** How do architectural components, fusion mechanisms, design choices, and semantic robustness contribute to TARL-TSC’s performance?

5.5.2. RQ1: Performance Benchmarking Across Traffic Scenarios

We evaluate whether TARL-TSC maintains competitive control performance under standard conditions while preparing the network for potential disruptions. We compare TARL-TSC against baseline methods under three scenarios defined in Section 5.2. Results are reported in Table 5. Under normal conditions, TARL-AGM achieves 869.1 throughput, 176.4 seconds ATT, and 4.78 AQL, improving over MPLight by 2.9 percent in throughput, 2.8 percent in ATT, and 1.8 percent in AQL. TARL-ATT shows competitive performance while maintaining higher real-time efficiency. As traffic conditions shift from normal flow to higher disruption levels, TARL-TSC maintains stable performance. TARL-AGM achieves 866.8 throughput under single-event and 864.5 under multi-event conditions, corresponding to only 0.3 percent and 0.5 percent degradation. TARL-ATT exhibits similar behavior with 859.1 and 857.2 throughput, indicating marginal variation across scenarios.

Table 5: Performance evaluation across the three traffic scenarios. Best and second-best results in **bold** and underline.

Method	Scenario 1 (Normal Flow)				Scenario 2 (Single Event)				Scenario 3 (Multi-Event)			
	Thr.	ATT (s)	AQL	AWT (s)	Thr.	ATT (s)	AQL	AWT (s)	Thr.	ATT (s)	AQL	AWT (s)
<i>Traditional Methods</i>												
Random	652	235.2	8.45	0.612	645	237.8	8.78	0.625	638	240.5	9.12	0.641
Fixed-Time	768	198.5	6.89	0.523	765	201.2	7.15	0.538	762	203.9	7.43	0.552
MaxPressure	820	187.3	5.92	0.489	815	189.8	6.18	0.501	810	192.4	6.45	0.515
<i>Deep RL Methods</i>												
PressLight	840	180.7	5.23	0.467	835	182.5	5.48	0.478	830	184.2	5.74	0.491
CoLight	842	182.9	4.98	0.442	838	184.5	5.21	0.453	835	186.1	5.46	0.465
MPLight	845	181.4	4.87	0.438	842	183.4	5.10	0.449	838	185.0	5.35	0.461
<i>Our Methods</i>												
TARL-ATT	<u>861.8</u>	<u>178.4</u>	4.92	0.448	<u>859.1</u>	<u>178.9</u>	4.89	0.445	<u>857.2</u>	<u>179.3</u>	4.95	0.447
TARL-AGM	869.1	176.4	4.78	0.445	866.8	177.1	4.81	0.443	864.5	177.8	4.85	0.445

Across all scenarios, TARL-TSC maintains stable performance as disruption severity increases. TARL-AGM achieves the highest throughput of 869.1, 866.8, and 864.5 across normal, single-event, and multi-event scenarios, respectively, indicating that performance gains are preserved under varying traffic conditions. This consistent performance implies that the proposed fusion mechanisms preserve control quality while incorporating semantic information without introducing instability. Figure 7 provides a visual comparison across evaluation metrics. TARL-TSC maintains performance advantages across different traffic scenarios relative to traditional and deep RL baselines, while exhibiting consistent behavior as conditions change.

These results shows that TARL-TSC achieves strong and stable performance across different traffic scenarios. Having established baseline performance, we next examine how the framework maintains this performance under traffic disruptions.

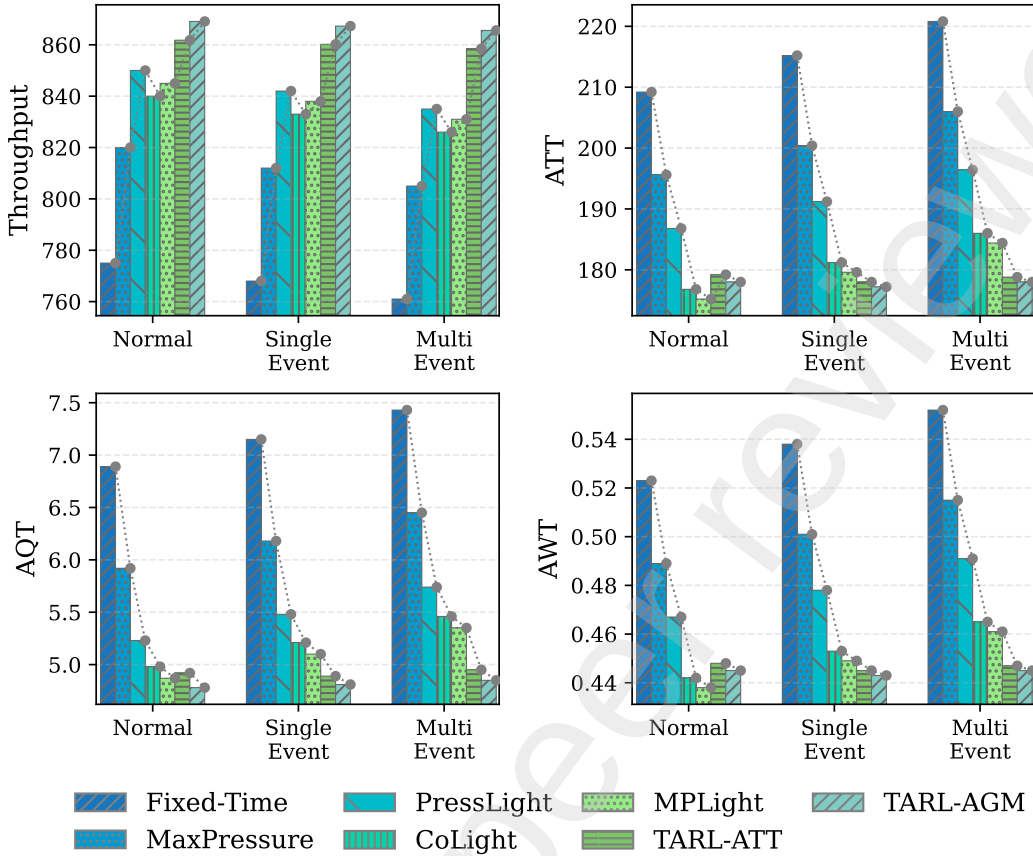


Figure 7: Performance comparison across traffic scenarios.

5.5.3. RQ2: Robustness to Event-Driven Traffic Disruptions

In this experiment, we evaluate the robustness of TARL-TSC under traffic disruptions. Specifically, we examine how the model behaves when traffic conditions deviate from nominal patterns due to events and when event information is delayed. We first assess robustness under six event types presented in Table 3 with increasing severity. Performance degradation is measured as the percentage decrease in throughput relative to the no-event baseline.

Figure 8 presents performance degradation across all methods and event types. TARL-AGM maintains degradation between 0.3 and 0.9 percent across all event severities, while TARL-ATT ranges between 0.4 and 1.1 percent. In comparison, MPLight exhibits larger degradation between 0.8 and 2.4 percent, and traditional methods degrade by up to 3.7 percent.

As event severity increases, TARL-TSC maintains low degradation, indicating stable behavior under non-recurrent traffic conditions. Under the most severe accident scenario, TARL-AGM shows 0.9 percent degradation compared to 2.4 percent for MPLight ($p = 0.003$), corresponding to a 2.6 $\times$ difference in performance drop. This behavior can be attributed to the adaptive fusion mechanism, where semantic information provides additional context when traffic observations alone become insufficient. In such cases, the model dynamically adjusts its reliance on text features to localize and respond to disruptions more effectively.

To further assess practical robustness, we evaluate performance under delayed availability of event descriptions, ranging from instantaneous to 5 minutes. Results show gradual performance degradation as delay increases. A 30-second delay results in only 0.5 percent degradation (865.6 throughput), while a 1-minute delay leads to 1.2 percent degradation (859.2 throughput). Even at 3-minute delays, degradation remains limited to 3.5 percent (837.5 throughput). Under an extreme 5-minute delay, TARL-TSC achieves 809.4 throughput with 6.8 percent degradation, remaining competitive with MPLight (845) and above the sensor-only baseline (840). Figure 9 visualizes this trend.

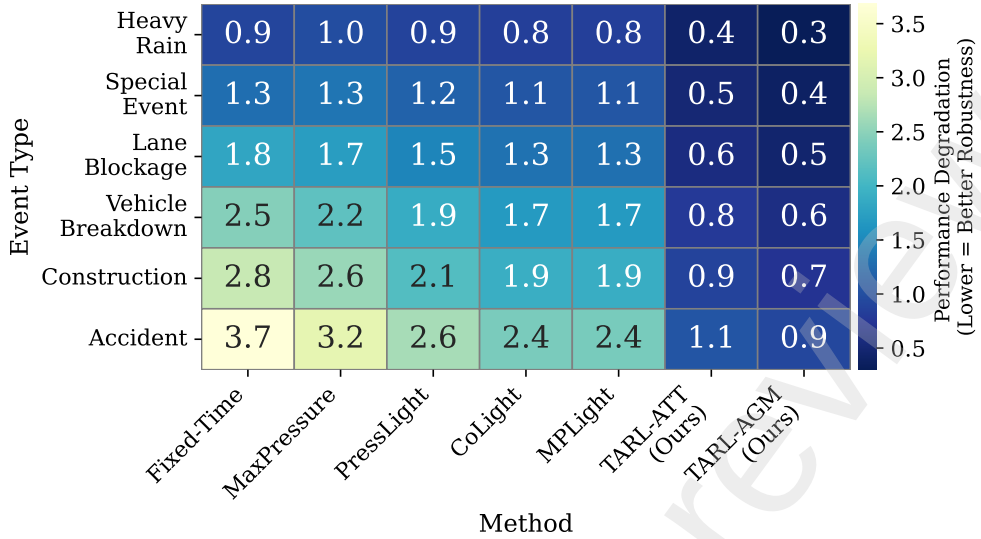


Figure 8: Performance degradation across event types.

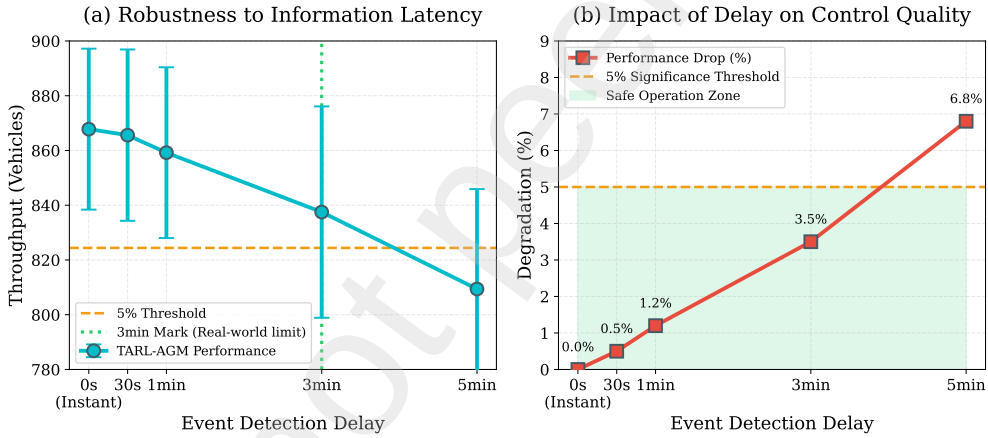


Figure 9: Performance under delayed event information.

These results show that TARL-TSC remains effective under realistic reporting delays, with minimal degradation for delays up to 3 minutes, consistent with real-world latency. TARL-TSC also shows strong robustness to event disruptions and delayed information, maintaining low degradation relative to baseline methods. This indicates that semantic event descriptions provide complementary information beyond traffic sensors, especially under non-recurrent conditions where disruptions cannot be inferred from local observations alone. By integrating contextual signals through language, the model is able to predict and adapt to disruptions more effectively than sensor-only approaches. We next examine whether this robustness generalizes across diverse urban environments.

5.5.4. RQ3: Cross-City Generalization to Unseen Traffic Networks

To evaluate the transferability of learned control policies, we assess cross-city generalization to unseen traffic networks with different topologies and scales. All learning-based models are trained and directly evaluated, without fine-tuning, on two unseen cities: Jinan and New York. These networks differ substantially in size, intersection density, and traffic demand patterns to provide a challenging generalization setting.

Quantitative throughput results are reported in Table 6. TARL-TSC achieves consistent improvements across all evaluated cities. In Jinan, TARL-AGM achieves 650.7 throughput, exceeding MPLight by 2.0 percent ($p = 0.038$). In Hangzhou, TARL-AGM attains 869.1, exceeding MPLight by 2.9 percent ($p = 0.032$). On New York, TARL-ATT

achieves 2625.5, exceeding MPLight by 1.2 percent ($p = 0.041$).

Table 6: Multi-city generalization without fine-tuning. Best and second-best results in **bold** and underline.

Method	Jinan (12 inter.)	Hangzhou (16 inter.)	New York (48 inter.)	Average
Fixed-Time	610	815	2540	1322
MaxPressure	620	825	2550	1332
PressLight	635	842	2590	1356
CoLight	632	840	2580	1351
MPLight	638	845	2595	1359
TARL-ATT	<u>645.9</u>	<u>861.8</u>	2625.5	1378
TARL-AGM	650.7	869.1	<u>2565.0</u>	<u>1362</u>

TARL-TSC maintains performance advantages across cities with different scales. Improvements are more pronounced on small- and medium-scale networks, while remaining consistent on larger networks. This suggests the learned policy captures transferable control patterns that generalize beyond the training environment. Figure 10 illustrates performance across cities. Despite substantial variation in network size (12 to 48 intersections), TARL-TSC exhibits stable behavior with limited variance. This indicates that the model does not rely on city-specific characteristics, but learns more general traffic control strategies.

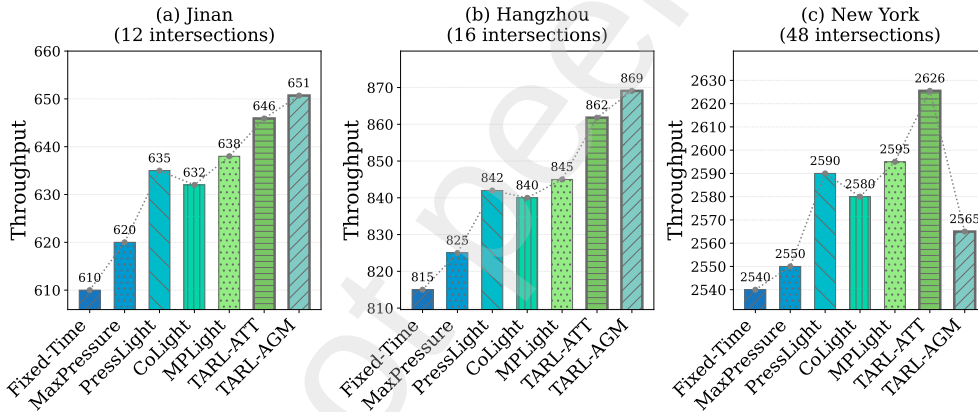


Figure 10: Cross-city generalization: 1.2-2.9% improvements over MPLight across 12-48 intersections without fine-tuning.

A comparison between fusion strategies reveals complementary behavior across network scales. TARL-AGM achieves stronger performance on small- and medium-scale networks, suggesting that adaptive gating balances traffic observations and semantic context in structured environments. In contrast, TARL-ATT performs better on the large New York network, where the increased spatial complexity may benefit from attention-based fusion that dynamically prioritizes relevant interactions across intersections.

These results demonstrate effective cross-city generalization to unseen traffic networks with different scales and topologies. TARL-TSC improves over baseline methods without requiring city-specific retraining, maintaining stable performance across diverse urban environments. Having established effectiveness and generalization, we now examine whether TARL-TSC meets the computational requirements for real-time deployment.

5.5.5. RQ4: Real-Time Efficiency for Practical Deployment

For practical deployment in urban traffic networks, signal control systems must make decisions within strict real-time constraints. This experiment evaluates computational efficiency across three dimensions: inference latency per decision, scalability across network sizes, and hardware resource requirements. Inference latency is measured under real-time deployment conditions, as detailed in Section 5.1.

The reported latency corresponds to per-step decision-making within the control loop. Event semantic embeddings are computed asynchronously upon event occurrence and cached before control execution.

Table 7 presents efficiency metrics for TARL-TSC variants across different network scales. In Hangzhou (16 intersections), TARL-AGM achieves 8.3 milliseconds inference time, while TARL-ATT requires 9.1 milliseconds. Both are substantially below the typical 1-second control cycle, which satisfies real-time deployment constraints.

Table 7: Real-time efficiency.

Network	Intersections	Method	Inference (ms)	Memory (MB)
Jinan	12	TARL-ATT	7.2	1,850
		TARL-AGM	6.8	1,720
Hangzhou	16	TARL-ATT	9.1	2,340
		TARL-AGM	8.3	2,180
New York	48	TARL-ATT	18.7	5,920
		TARL-AGM	22.4	5,680

Inference time increases approximately linearly with network size. Jinan (12 intersections) requires 6.8 milliseconds, while New York (48 intersections) requires 22.4 milliseconds. This scaling trend indicates TARL-TSC maintains sub-25ms inference even for larger networks, suggesting practical applicability to real-world deployments with moderate to large intersection counts.

The efficiency of TARL-TSC is attributed to its architectural design. Semantic information is incorporated during training through learned fusion, avoiding repeated language model inference through cached semantic representations. As a result, online decision-making relies only on lightweight neural components, enabling low-latency inference. Figure 11 illustrates the relationship between network size and inference time, showing TARL-TSC maintains stable and predictable latency growth as the number of intersections increases.

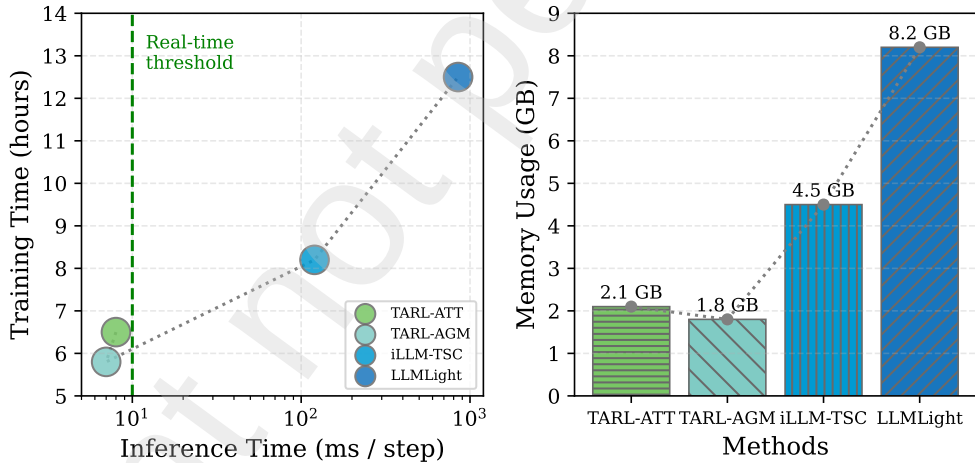


Figure 11: Inference time scaling.

Table 8 compares computational requirements across methods. TARL-TSC incurs higher training time and memory than lightweight RL baselines, but achieves improved control performance while maintaining real-time inference.

These results demonstrate practical feasibility for real-time traffic signal control. TARL-TSC maintains low-latency inference across network scales while achieving improved control performance, providing a balanced trade-off between efficiency and effectiveness. After we validated performance, robustness, generalization, and efficiency, we now conduct a comprehensive architectural analysis to understand the sources of these improvements.

5.5.6. RQ5: Architectural Contribution and Design Analysis

This section provides a comprehensive analysis of TARL-TSC’s architectural components and design choices to explain the performance gains observed in RQ1-RQ4. We examine four complementary aspects: (1) component contribution through ablation studies, (2) fusion mechanism behavior and positioning, (3) sensitivity to design

Table 8: Computational cost comparison.

Method	Training Time (hours)	Memory (MB)	Inference (ms)	Throughput (veh/ep)
PressLight	3.2	850	2.1	842
CoLight	4.5	1,240	4.3	840
MPLight	3.8	1,120	3.5	845
TARL-ATT	6.2	2,340	9.1	861.8
TARL-AGM	5.8	2,180	8.3	869.1

alternatives, and (4) robustness to semantic input variations. Together, these analyses validate architectural decisions and provide insights into the framework’s effectiveness.

Component Contribution Analysis. To quantify the contribution of each component, we conduct a structured ablation study in which individual modules are removed or modified while keeping all other settings unchanged. The evaluated variants include a traffic-only controller without semantic input, a model without spatial coordination, a variant using naive feature concatenation instead of learned fusion, and the two proposed fusion architectures. All variants are trained and evaluated under identical conditions.

Quantitative results are summarized in Table 9. The full model achieves 861.8 and 869.1 throughput for TARL-ATT and TARL-AGM, respectively. Removing text input reduces performance to 840.0 and 845.3, indicating that semantic information contributes even under nominal conditions. The variant without text also exhibits higher variance, suggesting reduced stability.

Table 9: Ablation study results.

Variant	TARL-ATT	TARL-AGM
Full Model	861.8 $\pm$ 37.6	869.1 $\pm$ 36.5
Without Text	840.0 $\pm$ 41.2	845.3 $\pm$ 39.8
Without Fusion (Concat)	855.2 $\pm$ 44.1	864.7 $\pm$ 38.2
Fine-tuned Encoder	859.1 $\pm$ 46.3	866.8 $\pm$ 40.1
Without GAT	842.5 $\pm$ 48.7	848.9 $\pm$ 45.3

Spatial coordination via GAT also contributes consistently. Removing GAT reduces throughput to 848.9, indicating that modeling inter-intersection dependencies is important. Similarly, replacing learned fusion with concatenation results in lower performance, confirming that adaptive integration is necessary. Figure 12 visually summarizes these component contributions, showing the relative impact of each architectural element on overall throughput performance.

The role of semantic information becomes more pronounced under non-recurrent conditions. In multi-event scenarios, the variant without text achieves 825.3 throughput, while the full model reaches 858.5. This gap highlights that textual context is particularly valuable when traffic observations alone are insufficient to capture disruption dynamics.

Fusion Mechanism Analysis. We further examine the effectiveness of the proposed fusion mechanisms. Learned fusion exceeds naive concatenation, with TARL-AGM achieving 864.7 throughput under concatenation compared to 869.1 in the full model. This indicates that dynamically weighting traffic and semantic features is more effective than treating them equally. Figure 13 illustrates the internal dynamics of the fusion modules. The results show that both gating and attention mechanisms adapt to traffic conditions. Under normal flow, traffic features dominate, while during event scenarios, the contribution of semantic information increases. This adaptive behavior explains the robustness observed in 5.5.3, where the model maintains stable performance under both traffic and semantic uncertainty.

To further understand these improvements, Figure 14 compares different fusion strategies. Early fusion (Fusion $\rightarrow$ GAT) achieves the best performance at 875 veh/ep, outperforming late fusion 841 veh/ep and the sensor-only baseline 826 veh/ep. This indicates that incorporating semantic information before spatial coordination allows for more effective context propagation and superior convergence stability.

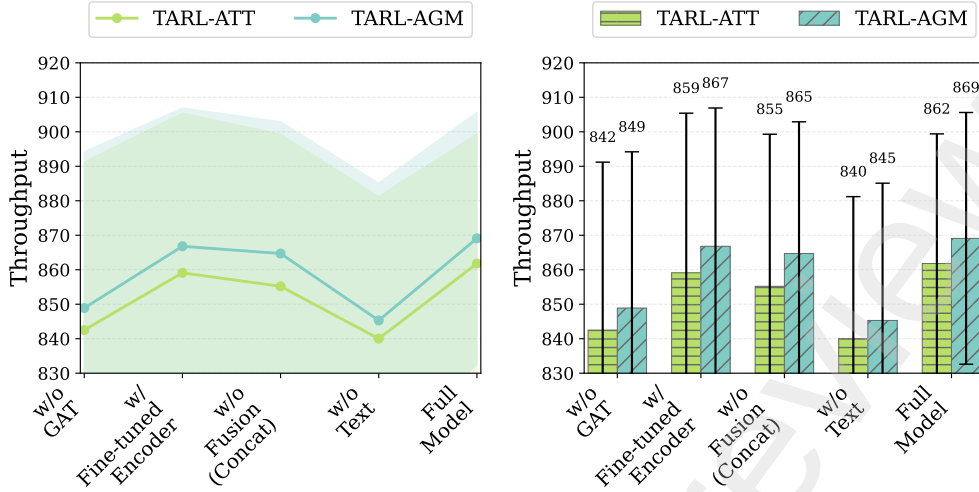


Figure 12: Component contribution analysis.

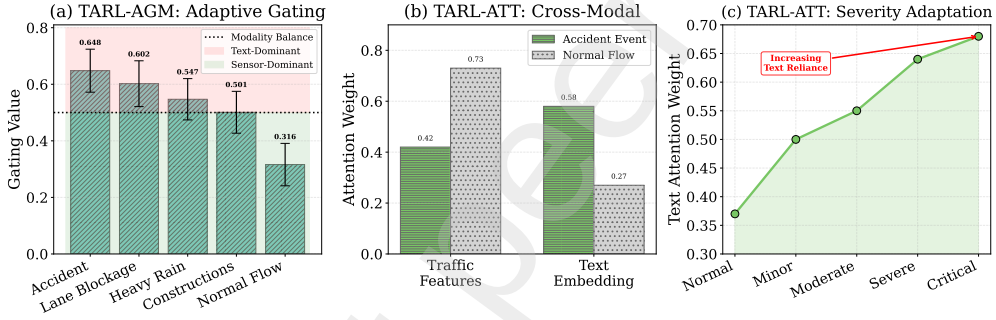


Figure 13: Adaptive gating behavior across traffic conditions.

Design Sensitivity Analysis. Beyond the core architectural components, we analyze the sensitivity of TARL-TSC to alternative design choices, including graph topology variants and temporal modeling strategies.

Table 10 compares different graph topology configurations. The fixed physical topology achieves stable performance with the lowest computational cost of 8.3 ms. More complex alternatives, such as attention-based or event-aware topologies, introduce additional overhead of 11.8-12.7 ms without consistent gains, indicating that the physical road network provides structural information.

Table 10: Graph topology comparison.

Topology Variant	Overall	Normal	Event	Inference (ms)
Fixed Physical	857.1 $\pm$ 32.0	866.4	866.1	8.3
Attention-Based	856.2 $\pm$ 31.3	870.5	857.3	12.7
Traffic-Flow-Based	848.4 $\pm$ 32.0	866.5	850.7	10.5
Event-Aware	854.8 $\pm$ 28.2	856.6	862.0	11.8

Table 11 evaluates temporal modeling strategies for event embeddings. Static embeddings achieve competitive performance with minimal computational overhead 8.3 ms. More sophisticated temporal models, such as LSTM-based temporal encoding, provide marginal improvements +0.63% but increase inference time to 15.7 ms, indicating that static embeddings provide sufficient semantic context for the evaluated event durations. We also compare different sentence encoders in terms of performance and efficiency.

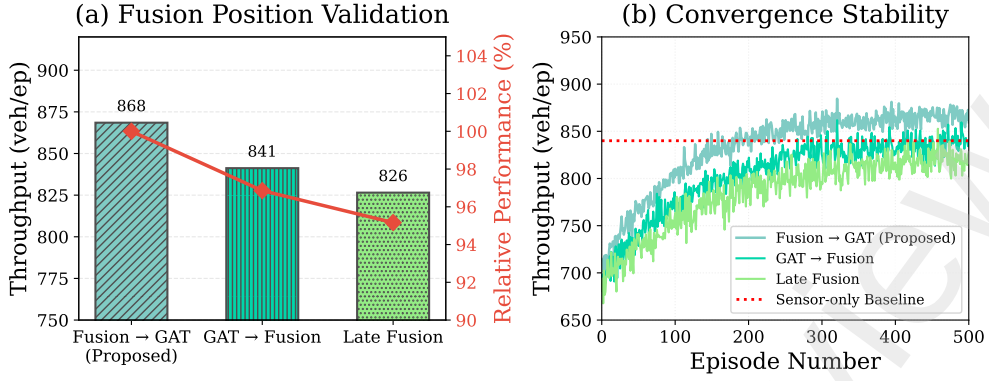


Figure 14: Fusion position comparison: early fusion exceeds other strategies.

Table 11: Temporal modeling comparison.

Approach	Throughput	Improvement	Adaptation	Inference (ms)
Static Embedding	850.5 ± 25.7	–	45	8.3
Updated Static	857.7 ± 24.7	+0.85%	25	8.5
Temporal Embedding	855.0 ± 27.4	+0.53%	30	9.1
LSTM Temporal	855.8 ± 25.6	+0.63%	20	15.7

As shown in Figure 15, all-mpnet-base-v2 achieves the best balance between text quality and computational cost. Smaller encoders reduce memory footprint but degrade performance, while larger models offer limited gains at a higher computational cost.

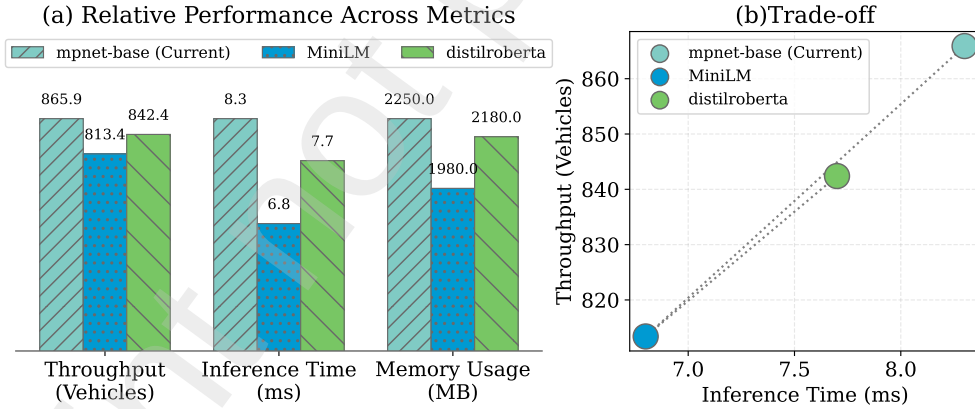


Figure 15: Comparison of text encoders.

Semantic Robustness Analysis. Beyond traffic-side robustness evaluated in 5.5.3, we examine the model’s robustness to variations in semantic input quality and language diversity. We first evaluate performance under different text quality conditions, including paraphrased descriptions, noisy inputs with irrelevant information, contradictory descriptions that conflict with sensor observations, and completely unrelated text. Table 12 summarizes the results.

The results show that TARL-TSC maintains stable performance under moderate variations as illustrated in Figure 16(a). Paraphrased descriptions lead to minimal degradation of 0.7%, indicating that the model does not rely on surface-level patterns. This suggests that the model does not rely on exact text matching or fixed embedding lookup, but instead captures underlying semantic representations across varied linguistic forms.

We further evaluate robustness under broader language diversity, testing TARL-TSC on event descriptions

Table 12: Performance under different text quality conditions.

Text Quality Variant	Throughput	Degradation
Baseline	868.3 $\pm$ 23.6	—
Paraphrased	862.3 $\pm$ 32.0	0.7%
Noisy	852.2 $\pm$ 27.8	1.9%
Contradictory	846.5 $\pm$ 32.9	2.5%
Irrelevant	830.9 $\pm$ 31.6	4.3%

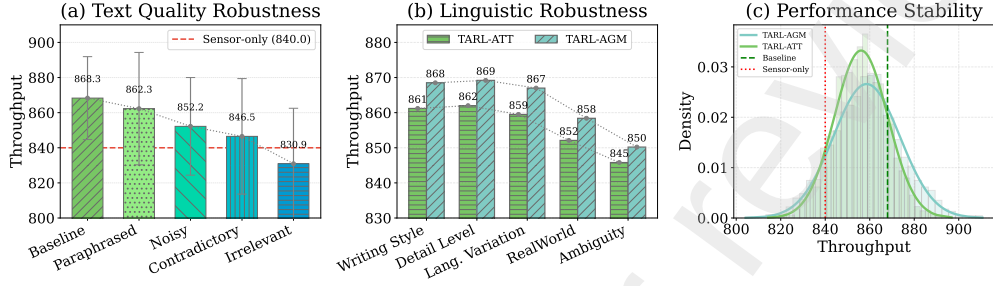


Figure 16: Semantic robustness analysis.

covering different writing styles, levels of detail, ambiguity levels, and real-world variations such as misspellings and abbreviations. Table 13 presents the results across these categories.

Table 13: Performance across language diversity categories.

Category	Throughput	Avg Degradation	Max Degradation
Writing Style	863.3 $\pm$ 24.9	0.70%	2.74%
Detail Level	863.3 $\pm$ 24.9	0.61%	1.96%
Language Variation	860.8 $\pm$ 23.6	0.53%	2.16%
Real World	851.4 $\pm$ 24.3	1.69%	3.69%
Ambiguity	848.4 $\pm$ 27.1	2.37%	5.44%
Overall Average	857.8 $\pm$ 25.6	1.13%	5.44%

To evaluate the framework’s generalization beyond fixed templates, we subjected TARL-TSC to 1,700 diverse event descriptions spanning five linguistic categories. As illustrated in Figure 16(b-c), performance remains stable across writing styles and language variations. While a minor gradual degradation in throughput is observed as semantic ambiguity and real-world noise increase, the model maintains a performance above the sensor-only baseline. This suggests that the framework captures underlying latent semantics rather than over-fitting to specific prompt structures.

These results show that TARL-TSC’s performance stems from the successful integration of semantic augmentation, spatial coordination, and adaptive multimodal fusion. While each component contributes to overall gains, the semantic stream is important when traffic events occur and sensor data is limited. By weighting these inputs, the framework maintains high throughput even under noise. TARL-TSC achieves a balance between computational efficiency and operational resilience, navigating both complex traffic conditions and real-world semantic ambiguity.

Across RQ1–RQ5, the results show that incorporating natural language descriptions of traffic events improves traffic signal control performance. Although the gains are modest under normal conditions, they become increasingly substantial under uncertainty, including event disruptions, delayed information, and semantic variation. Ablation and robustness analyses further demonstrate that these improvements come directly from the semantic information itself rather than from architectural complexity. By providing contextual cues that traffic sensors cannot capture, language enables more adaptive and resilient control policies.

6. Conclusion

This paper introduces TARL-TSC, a text-augmented reinforcement learning framework that integrates natural language event descriptions with traffic observations for adaptive signal control. The framework overcomes existing RL-based methods that rely on sensor data and struggle to incorporate semantic information about traffic disruptions. TARL-TSC employs two learned fusion mechanisms, adaptive gating and attention-based fusion, to dynamically integrate traffic observations with semantic event embeddings. The framework uses graph attention networks for spatial coordination and operates within a multi-agent RL setting where each intersection learns a decentralized control policy.

Experimental evaluation on three traffic networks shows consistent improvements. TARL-TSC achieves 869.1 throughput in Hangzhou, exceeding MPLight by 2.9 percent. Under event disruptions, TARL-TSC maintains degradation between 0.3 and 0.9 percent compared to 0.8 to 2.4 percent for MPLight. Cross-city evaluation shows improvements of 1.2 to 2.9 percent across networks with 12 to 48 intersections without fine-tuning. The framework achieves real-time inference at 8.3 milliseconds per decision, meeting deployment constraints.

Architectural analysis shows that each component contributes to performance: semantic augmentation provides robustness under disruptions, spatial coordination supports network-wide optimization, and learned fusion mechanisms adaptively weight modalities by traffic conditions. The framework is robust to traffic disruptions and semantic variations, including text quality degradation and language diversity. Future work may explore temporal modeling of evolving event descriptions, extension to heterogeneous intersection types, and deployment in real-world traffic management systems.

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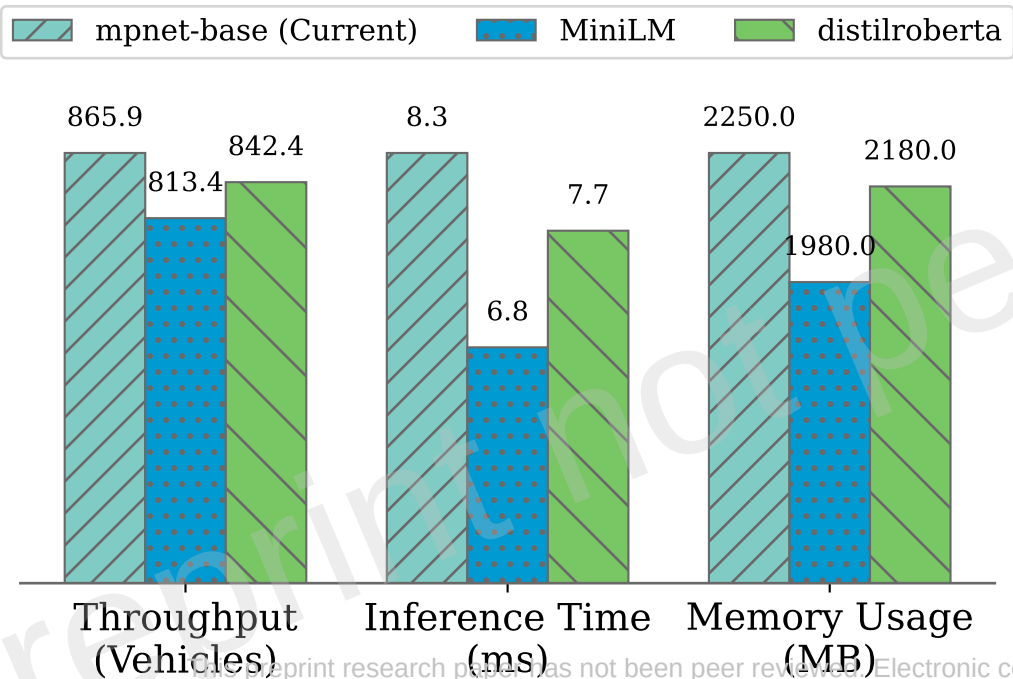
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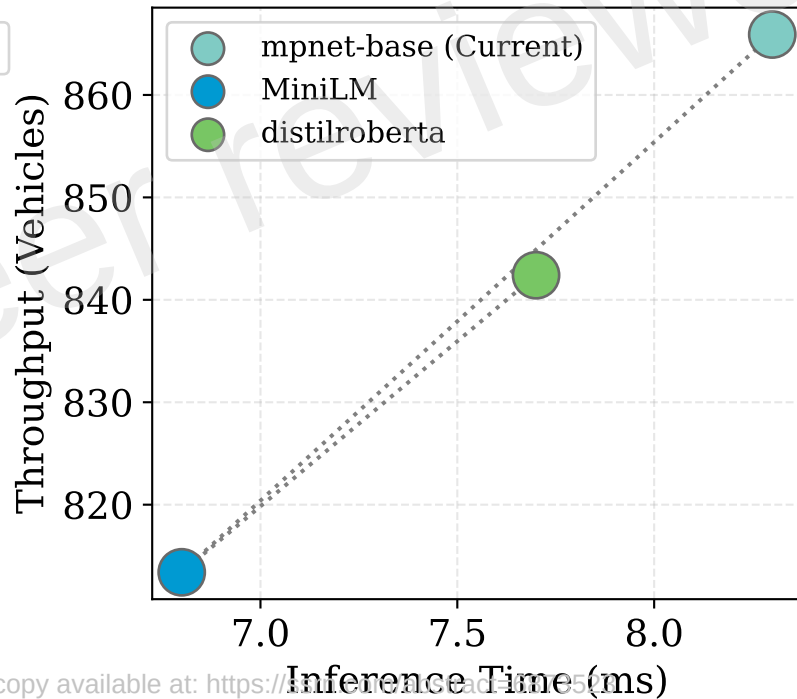
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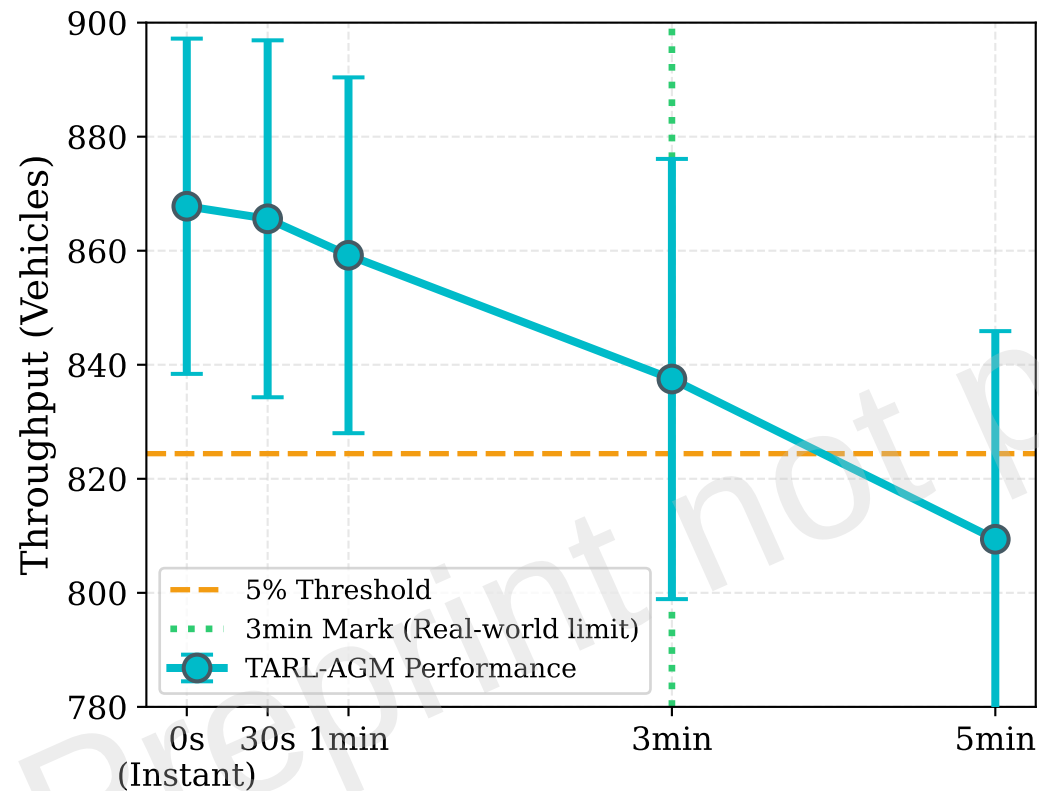
(a) Relative Performance Across Metrics



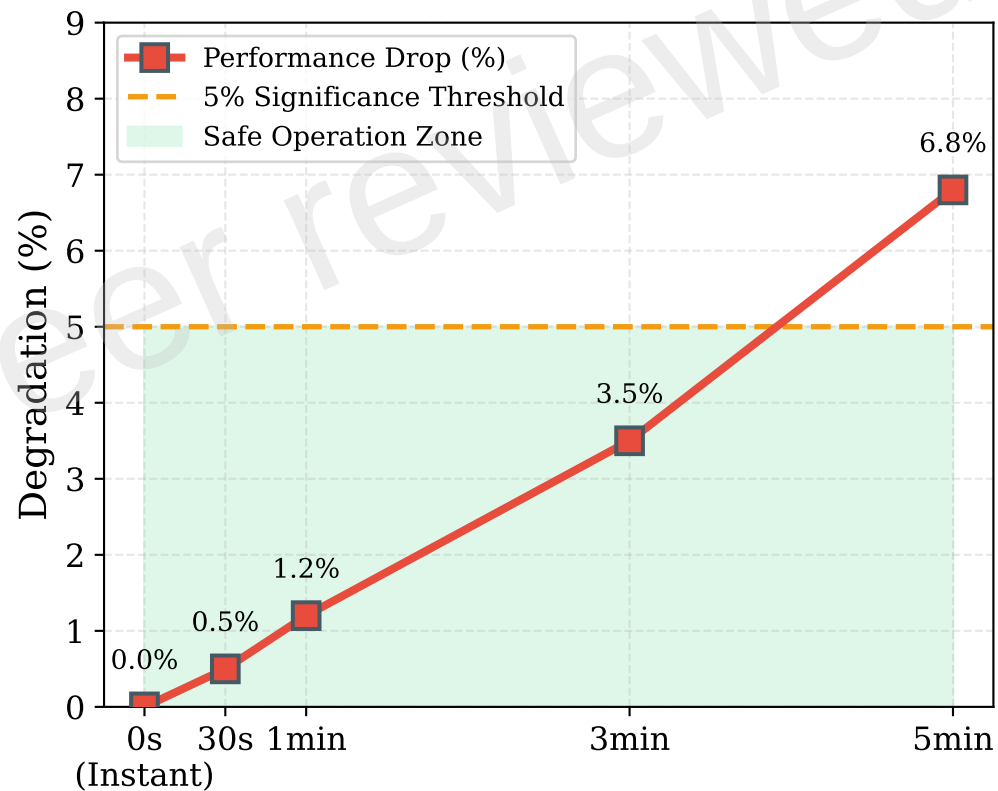
(b) Trade-off



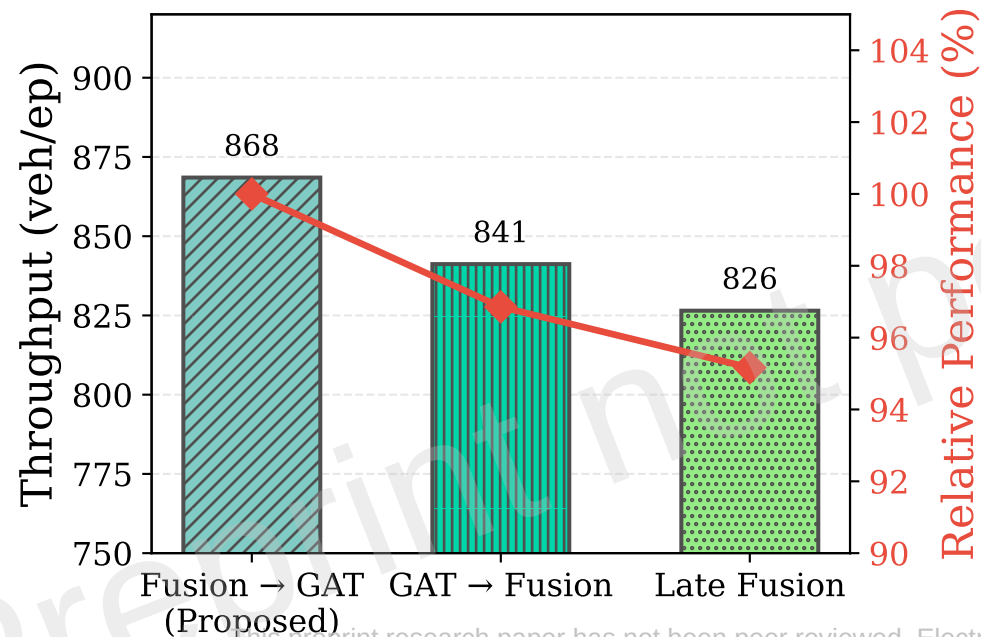
(a) Robustness to Information Latency



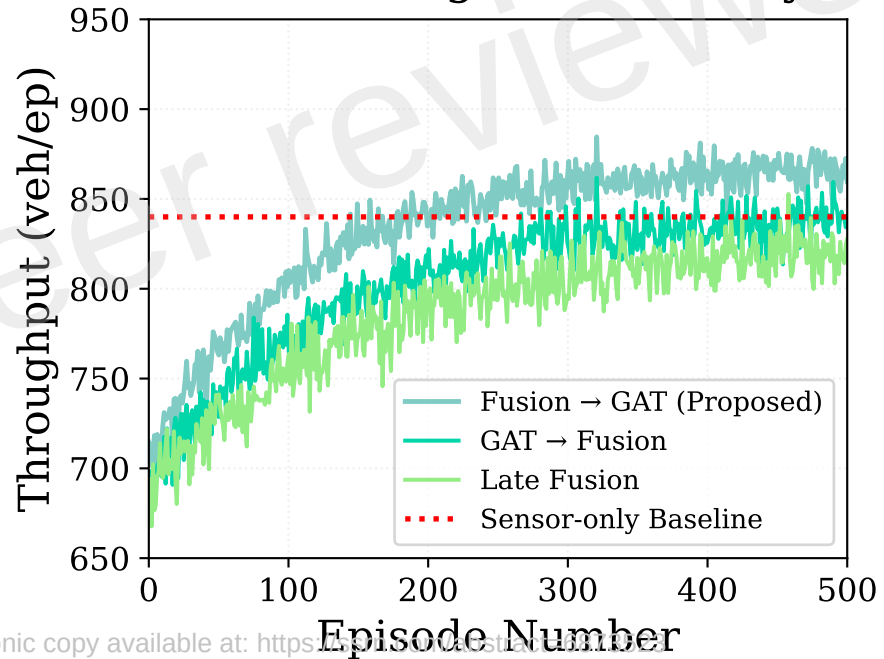
(b) Impact of Delay on Control Quality



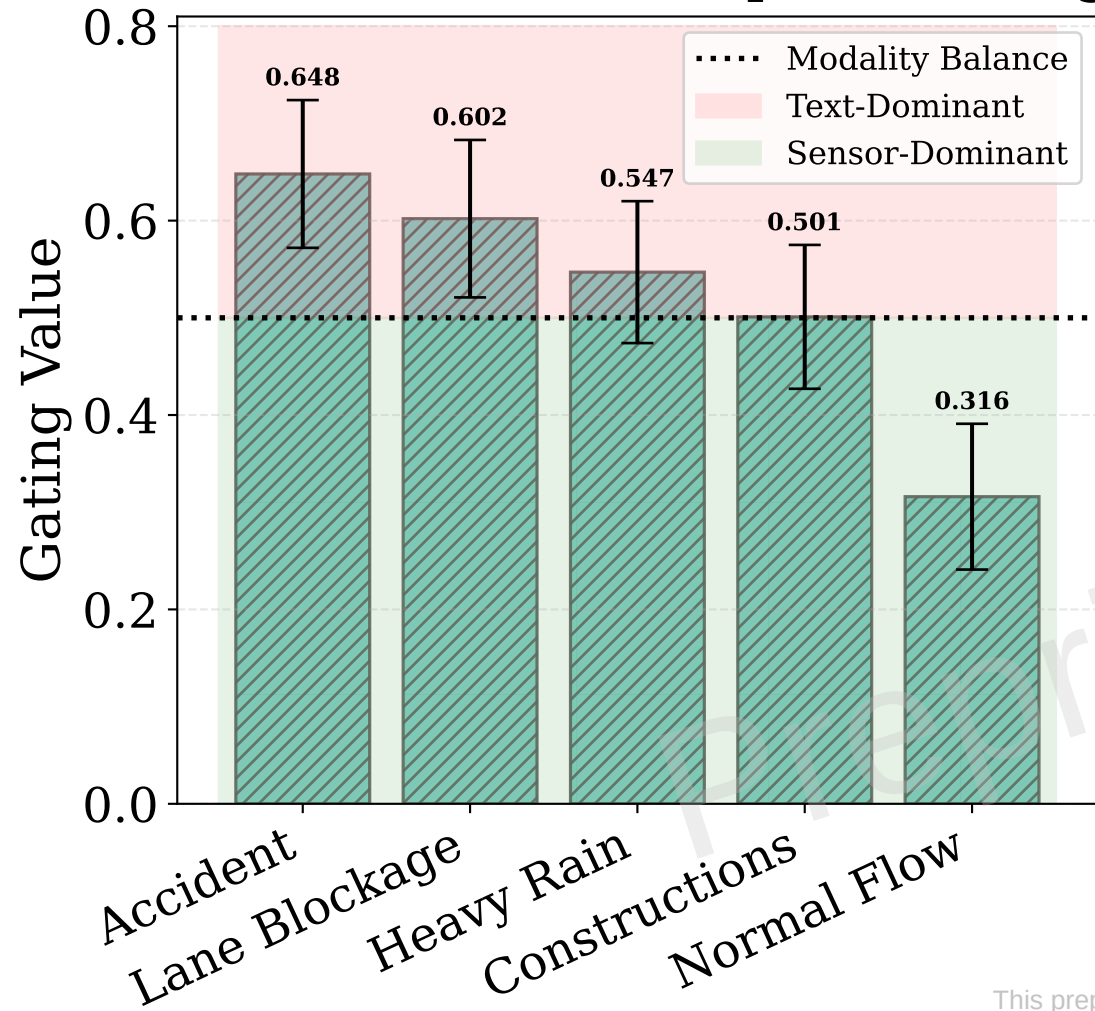
(a) Fusion Position Validation



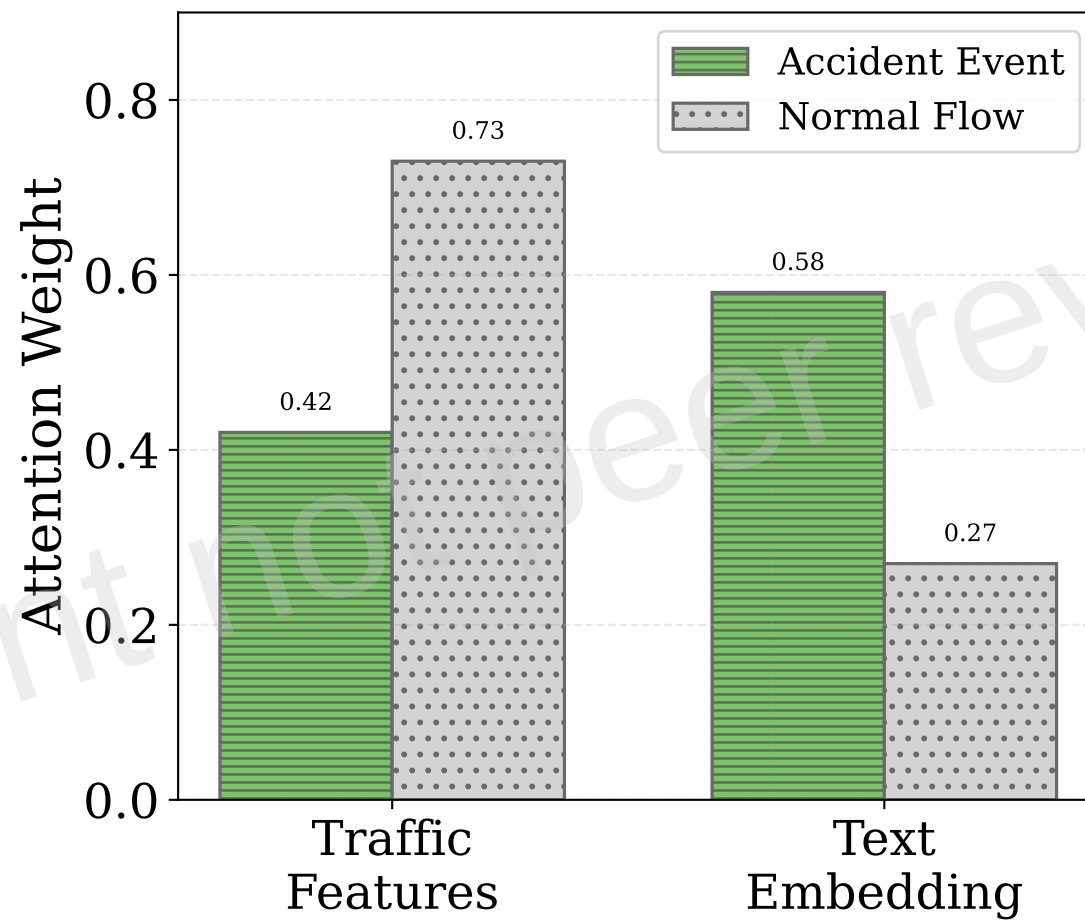
(b) Convergence Stability



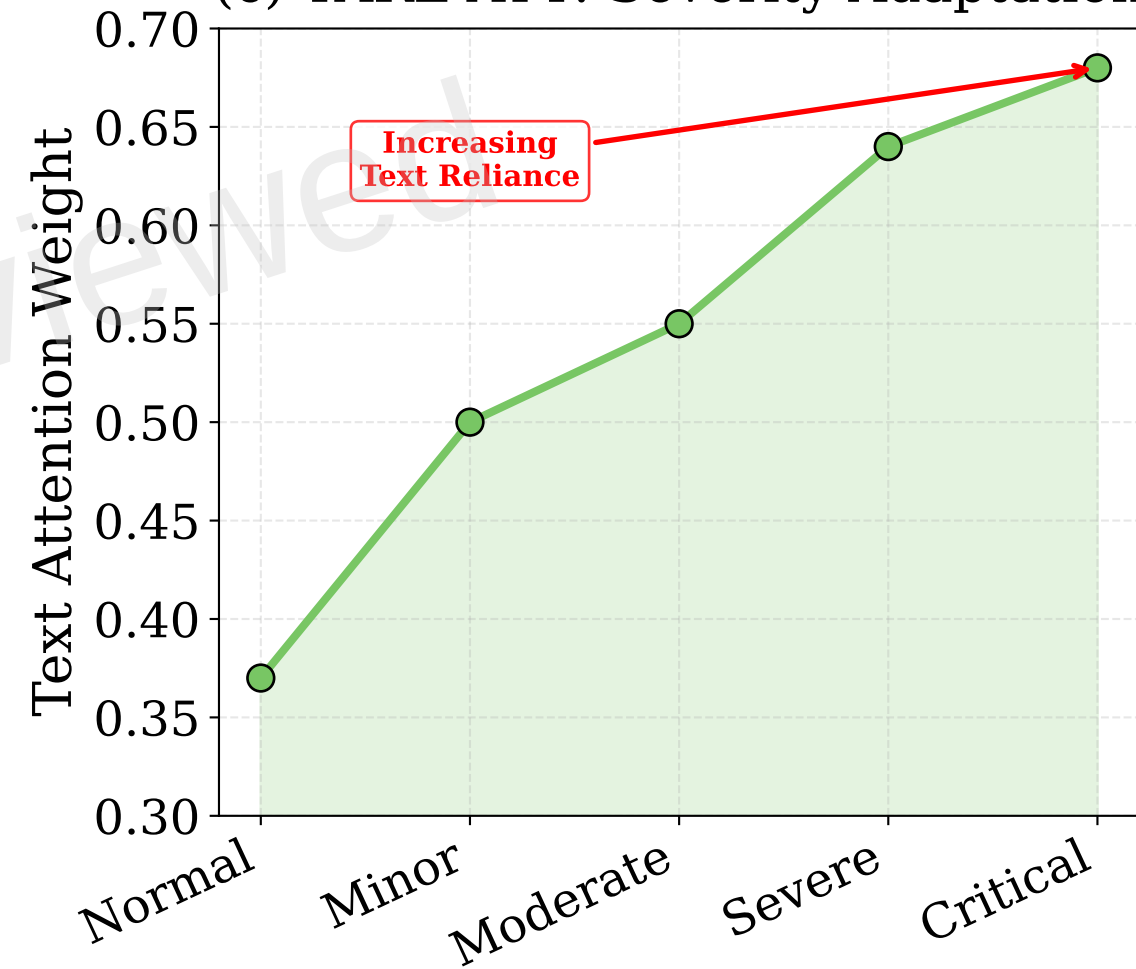
(a) TARL-AGM: Adaptive Gating

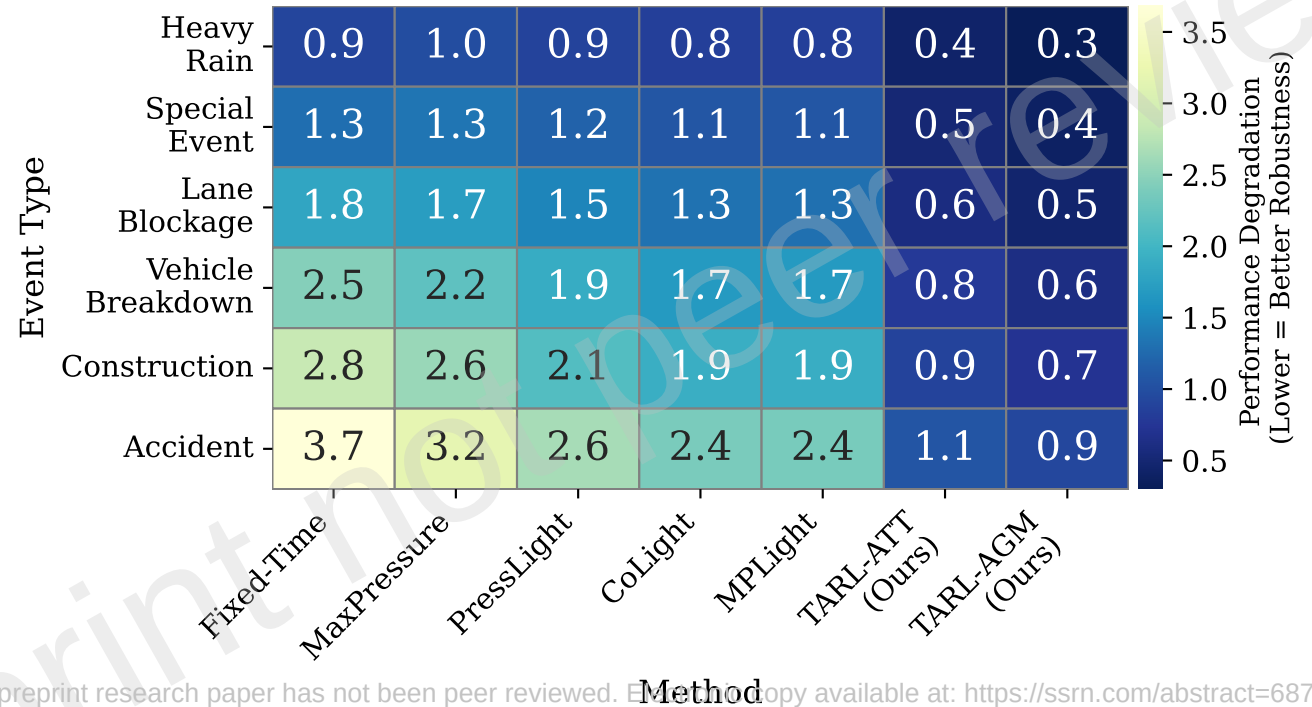


(b) TARL-ATT: Cross-Modal

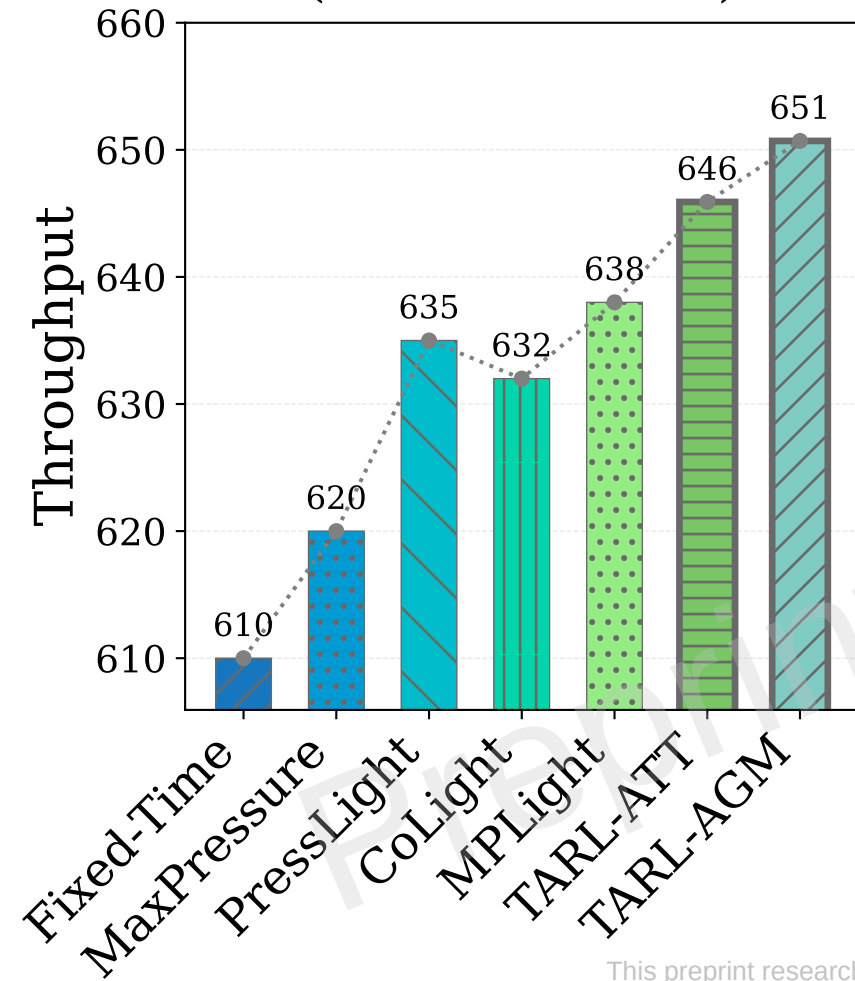


(c) TARL-ATT: Severity Adaptation

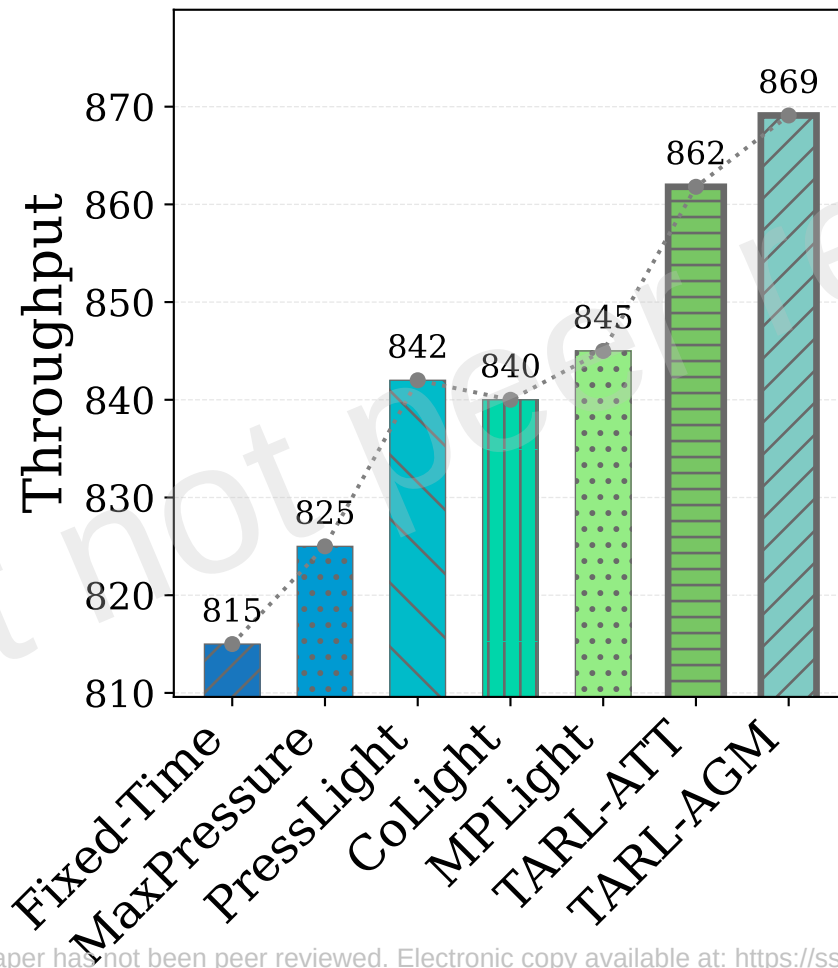




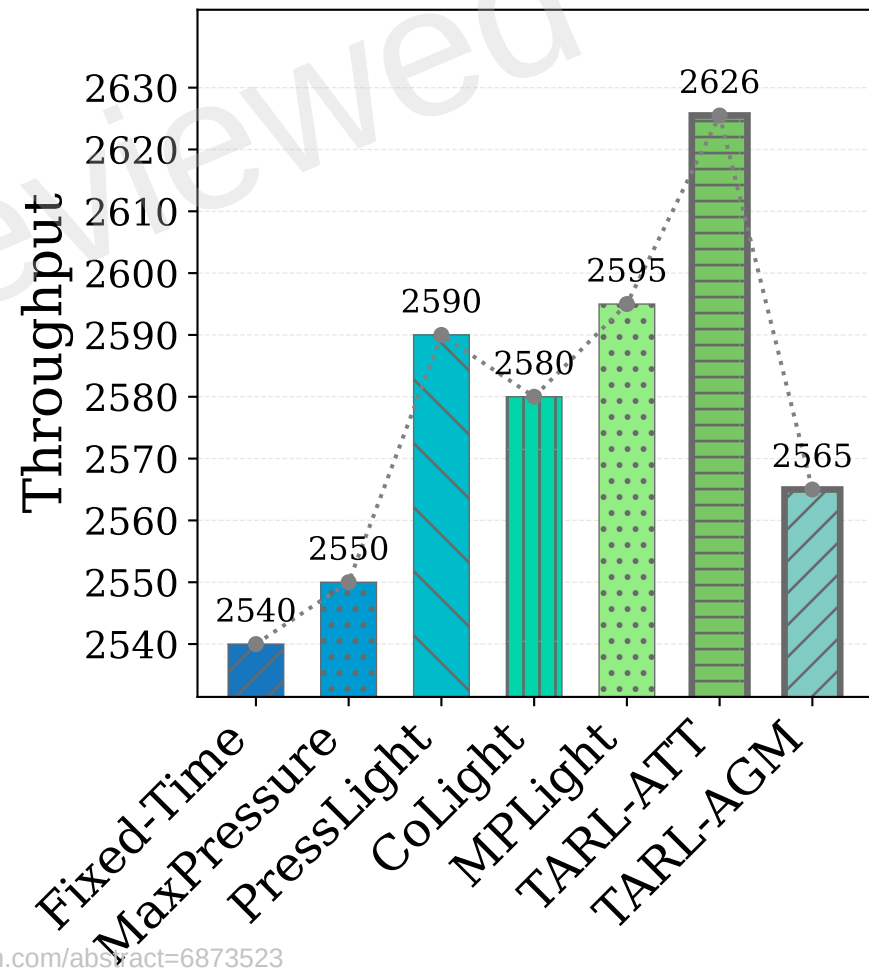
(a) Jinan
(12 intersections)

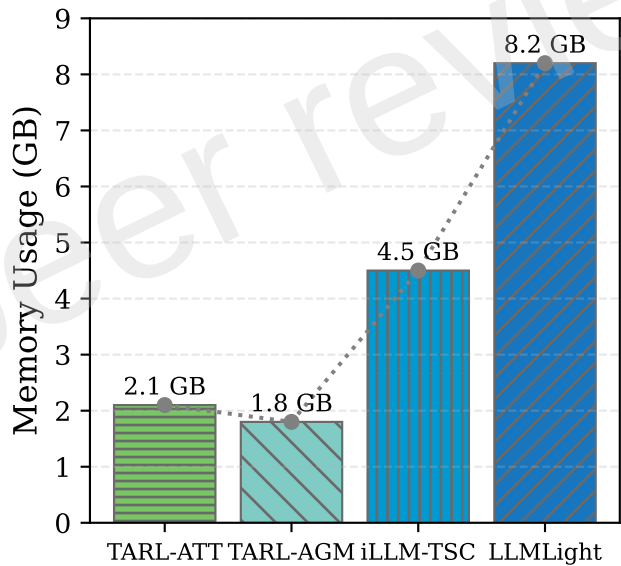
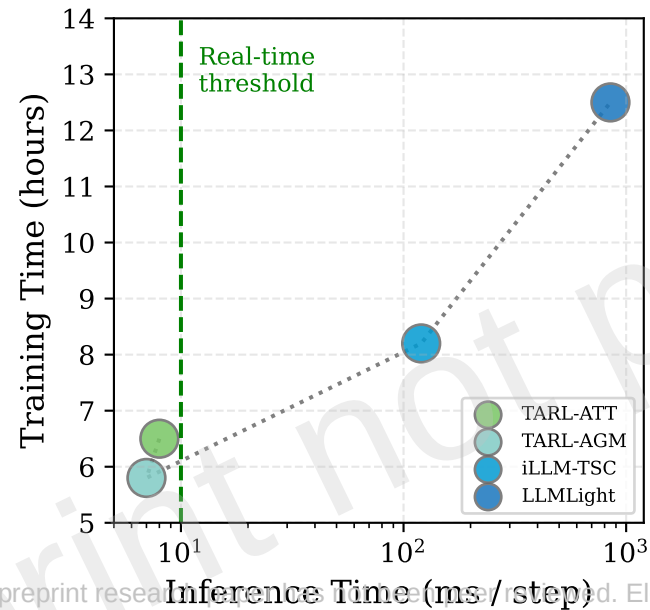


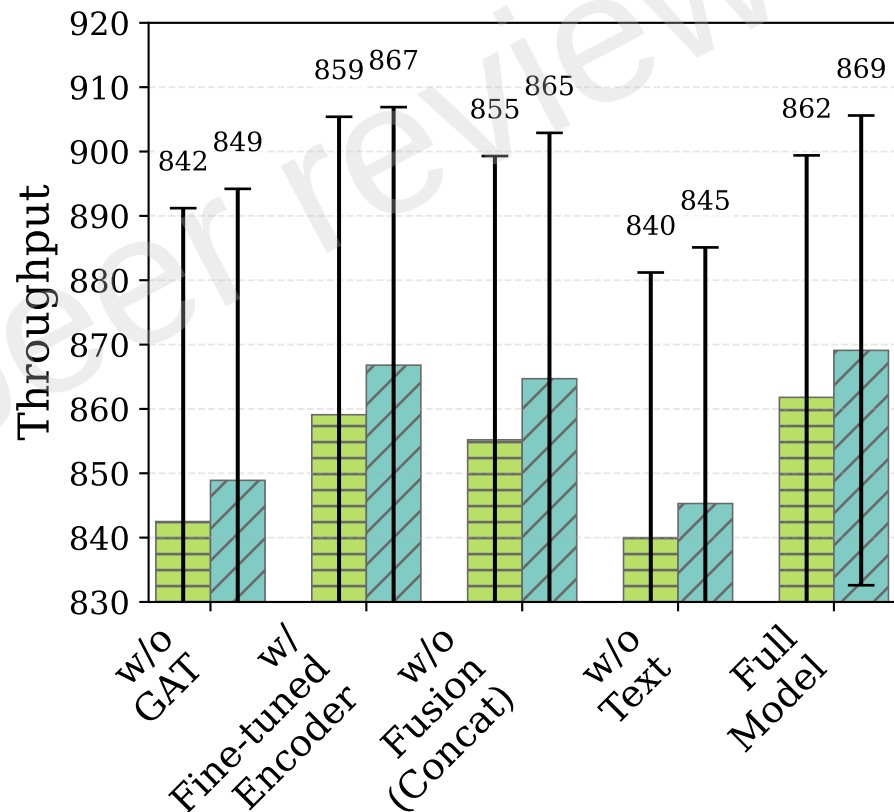
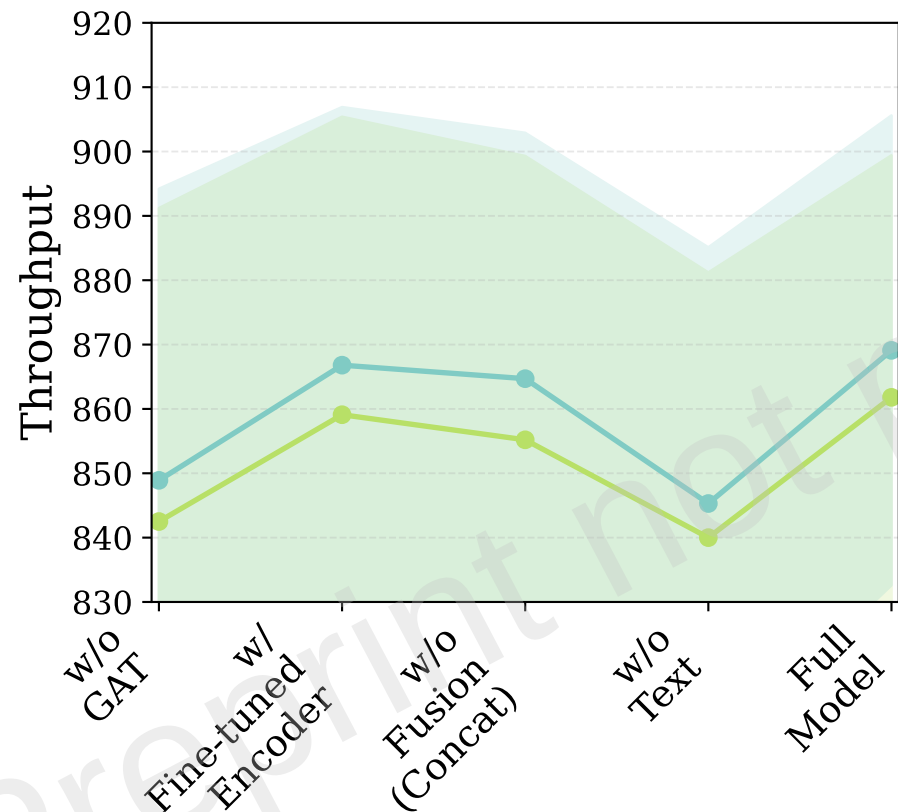
(b) Hangzhou
(16 intersections)



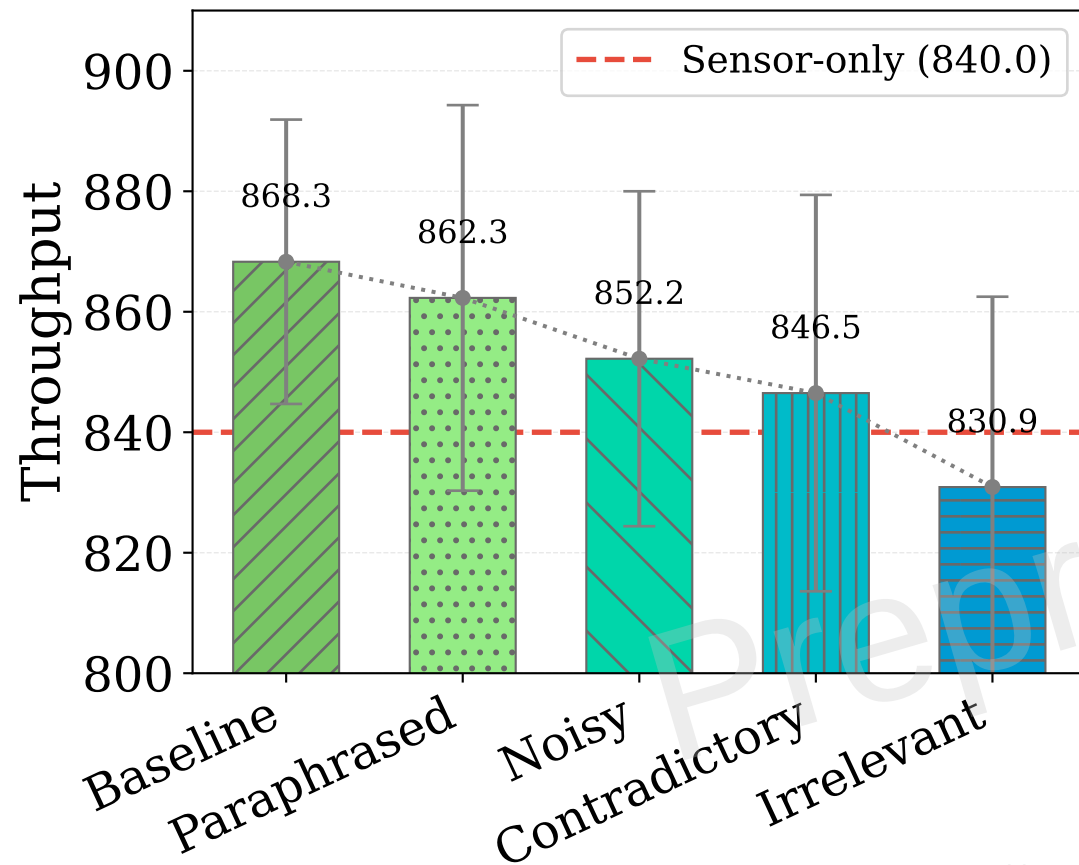
(c) New York
(48 intersections)



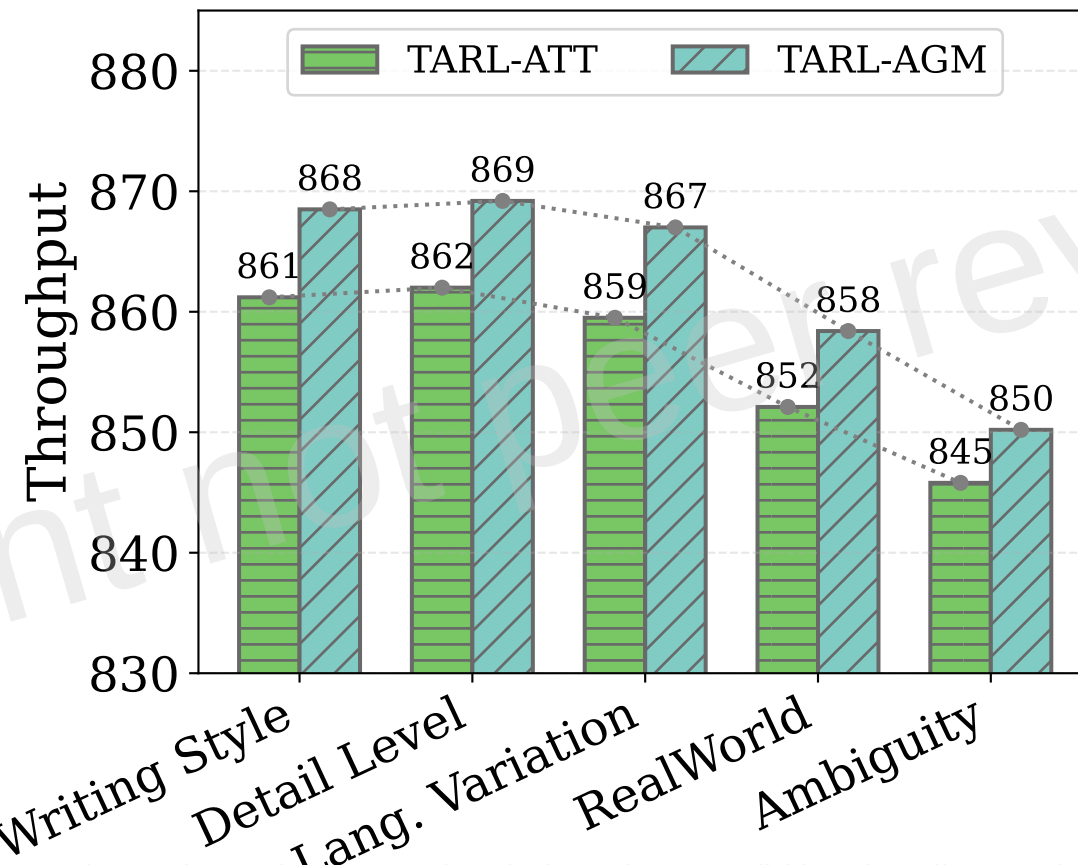




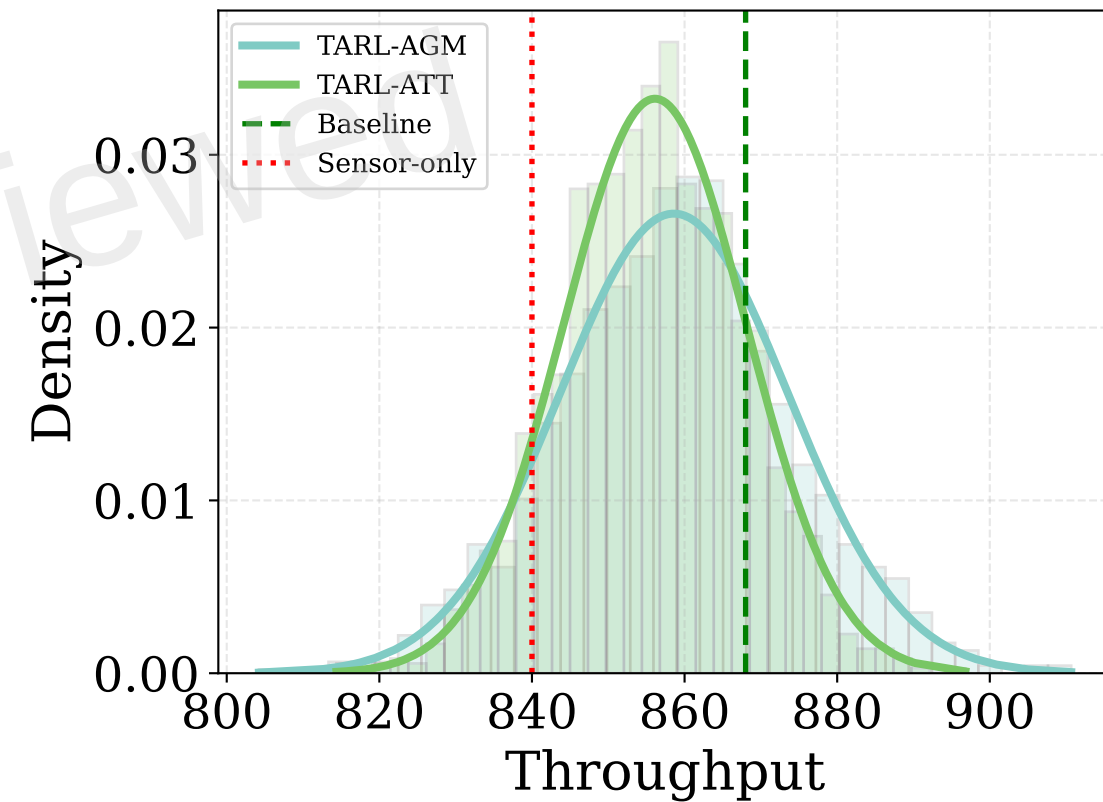
(a) Text Quality Robustness

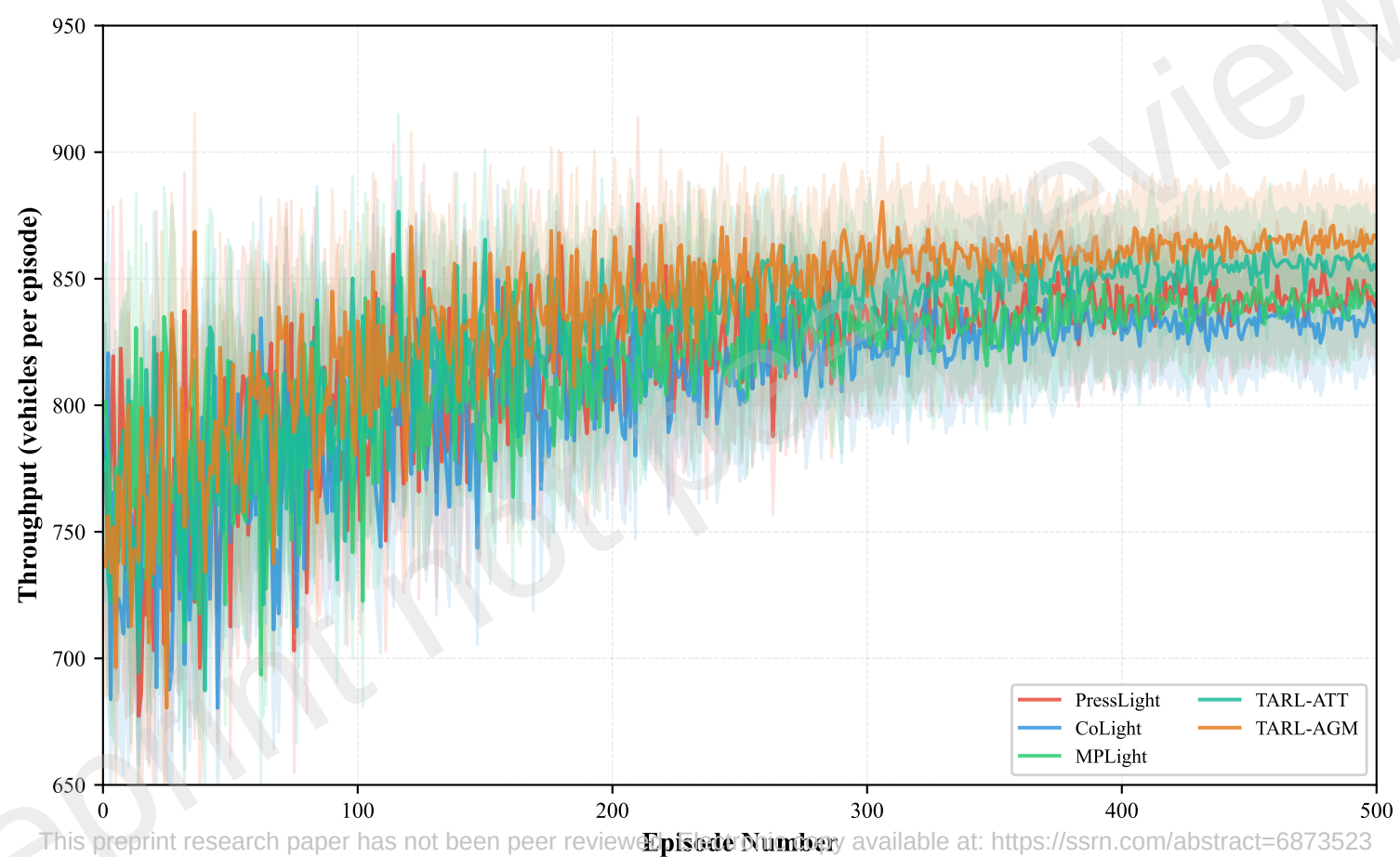


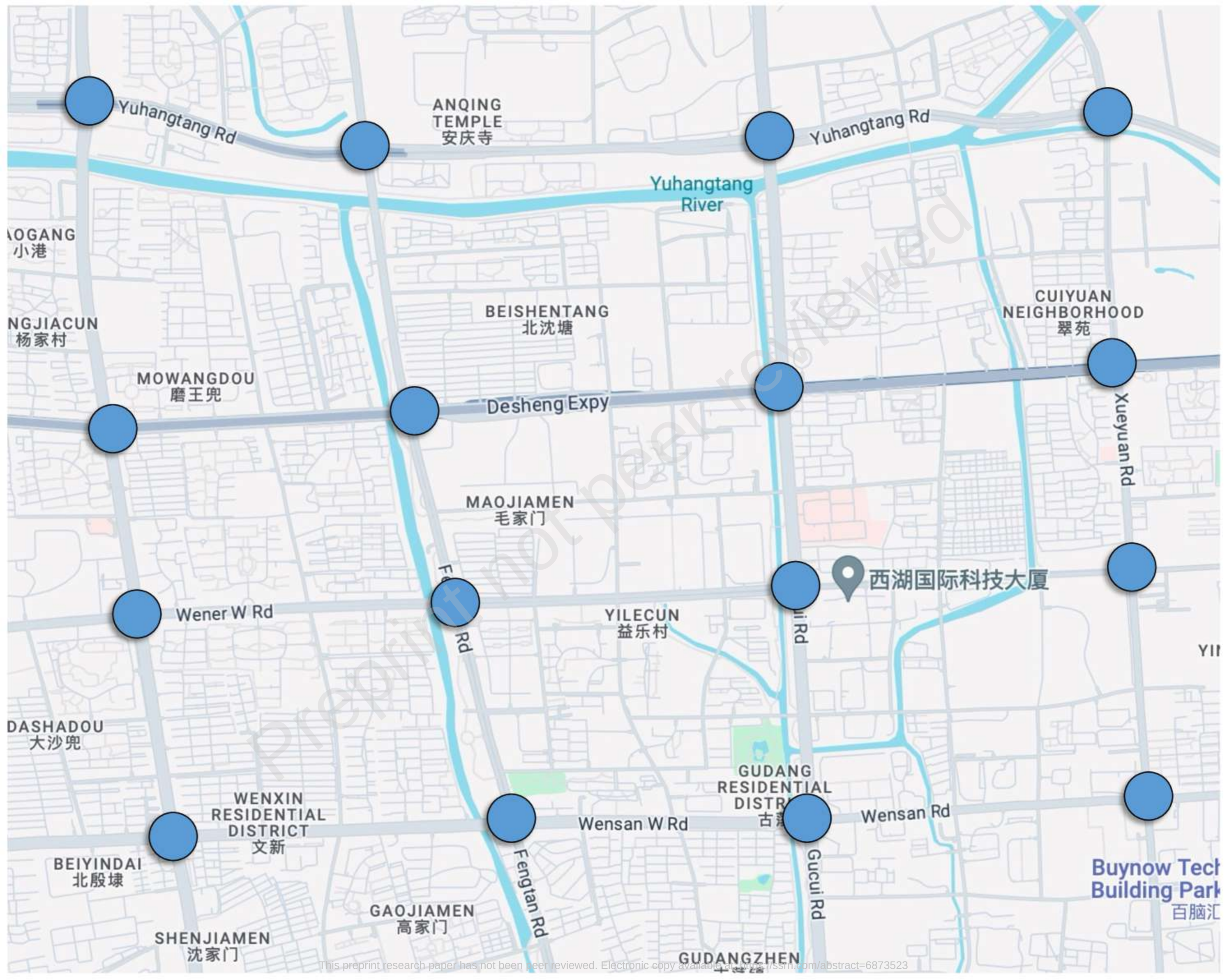
(b) Linguistic Robustness

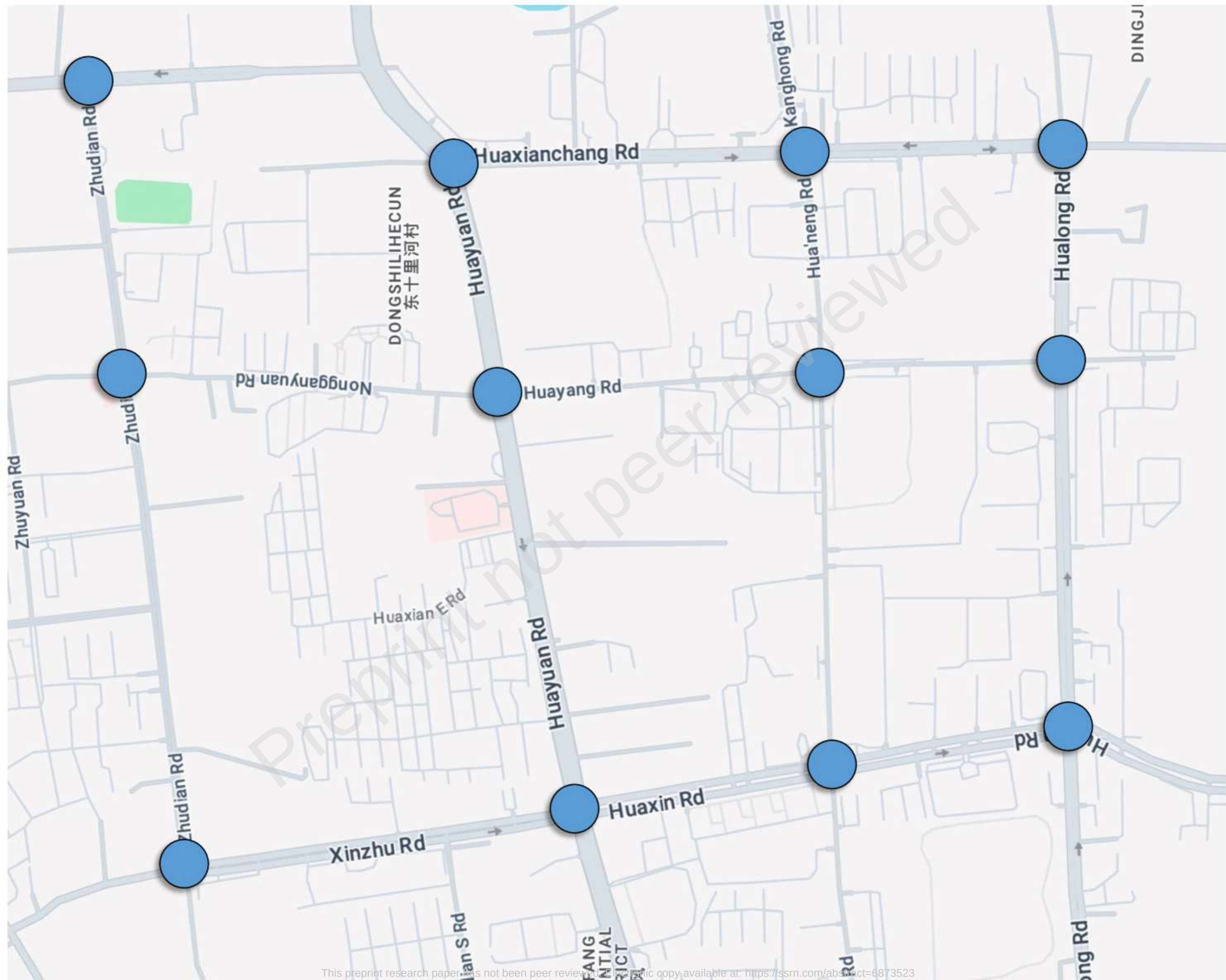


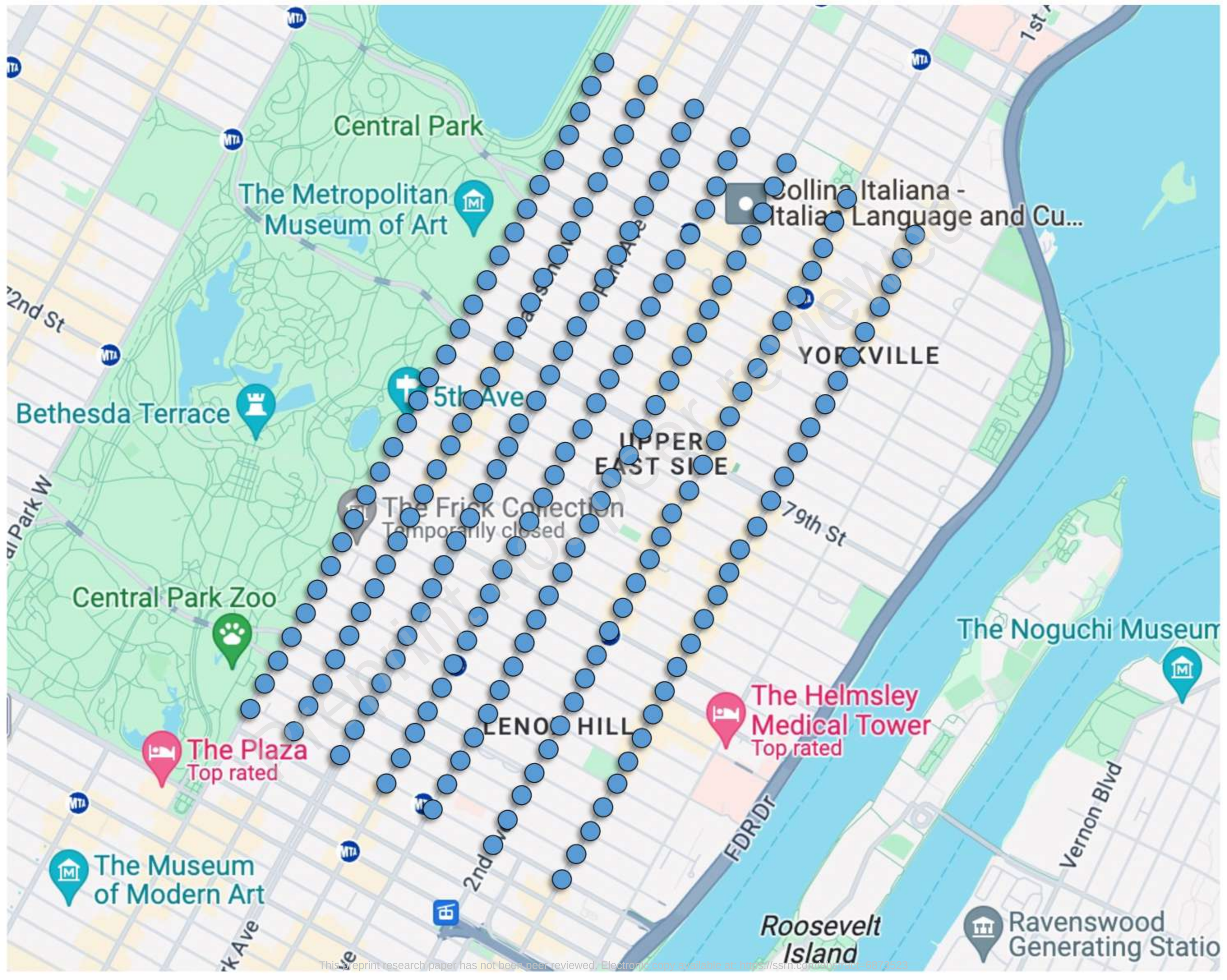
(c) Performance Stability

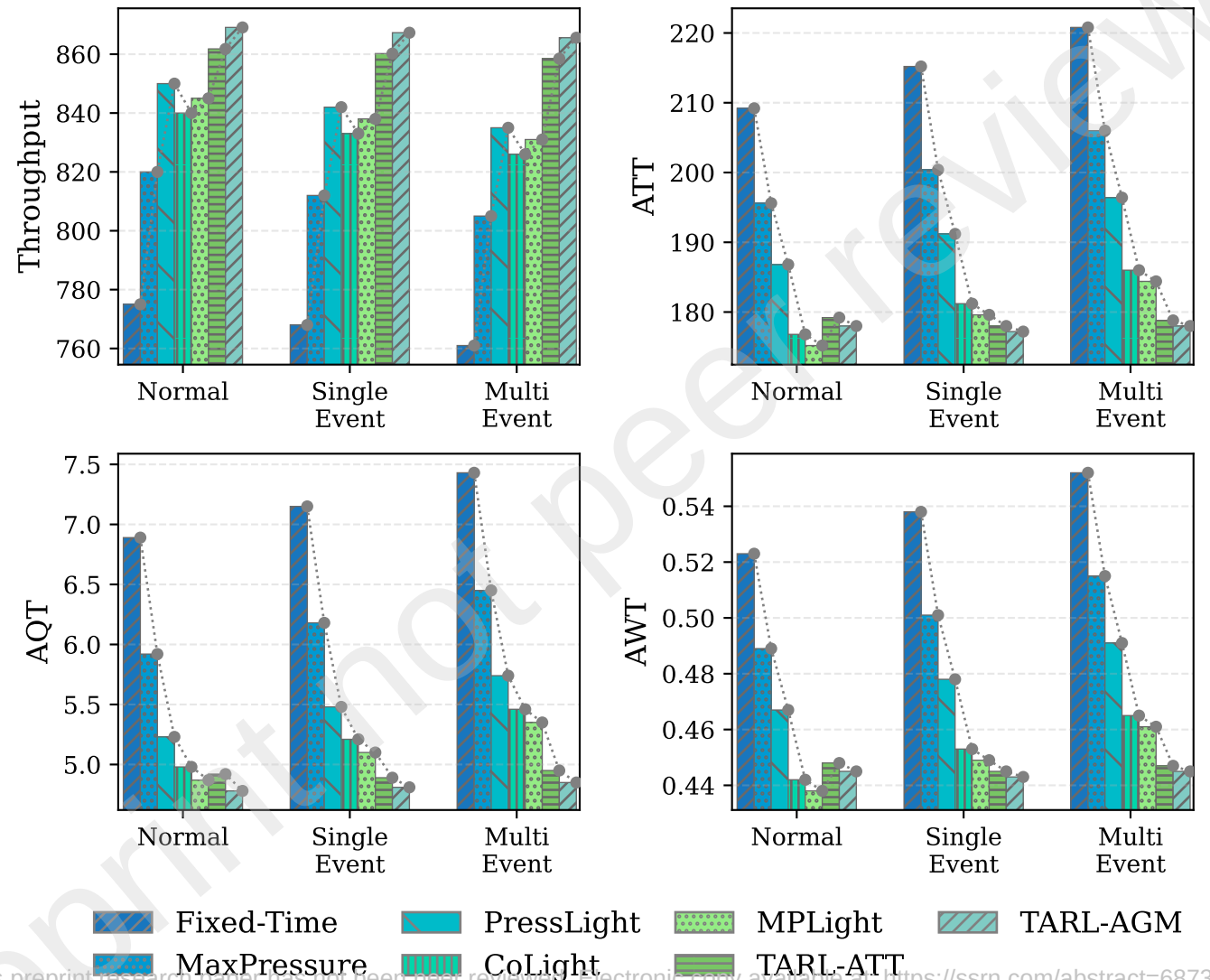


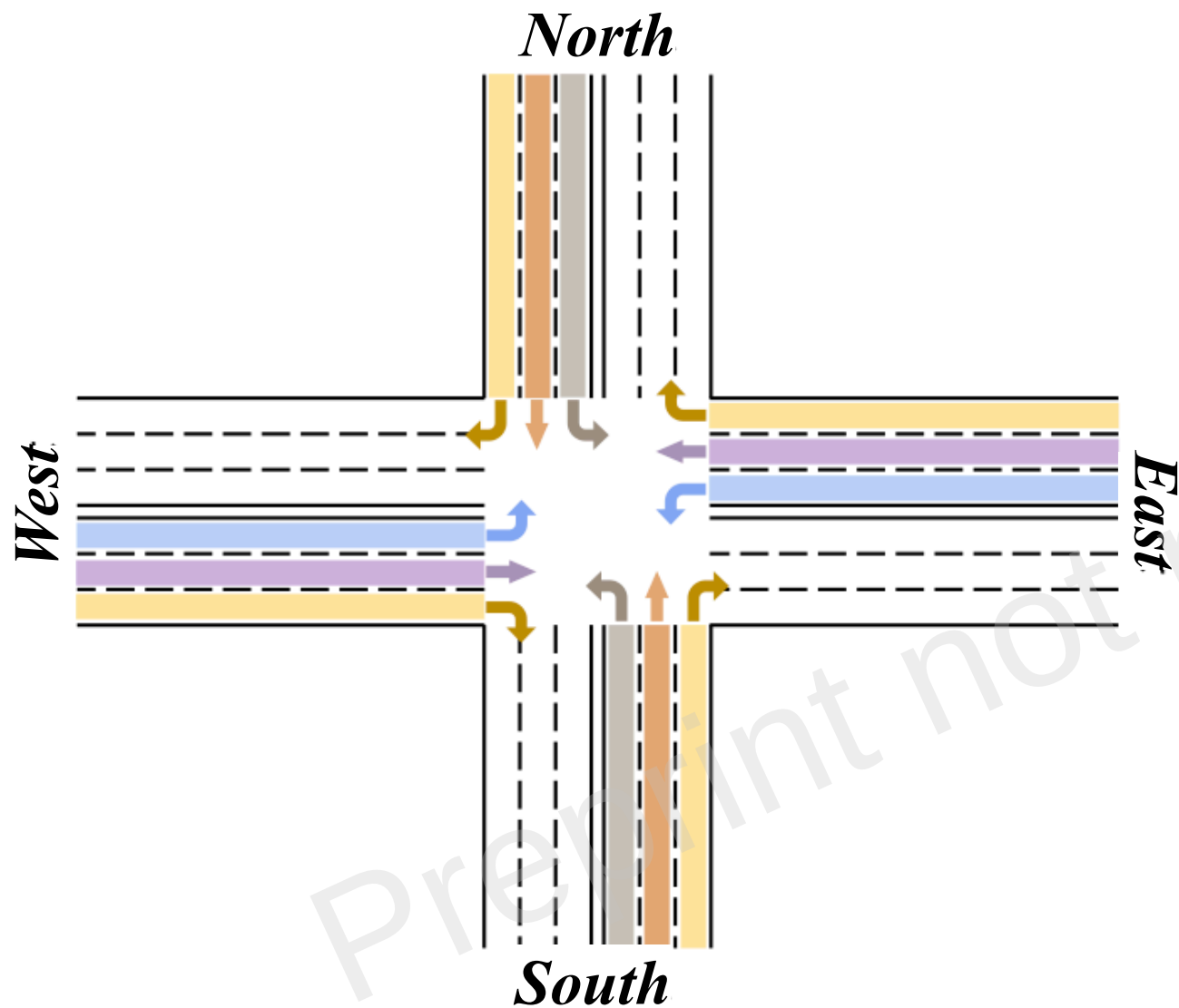




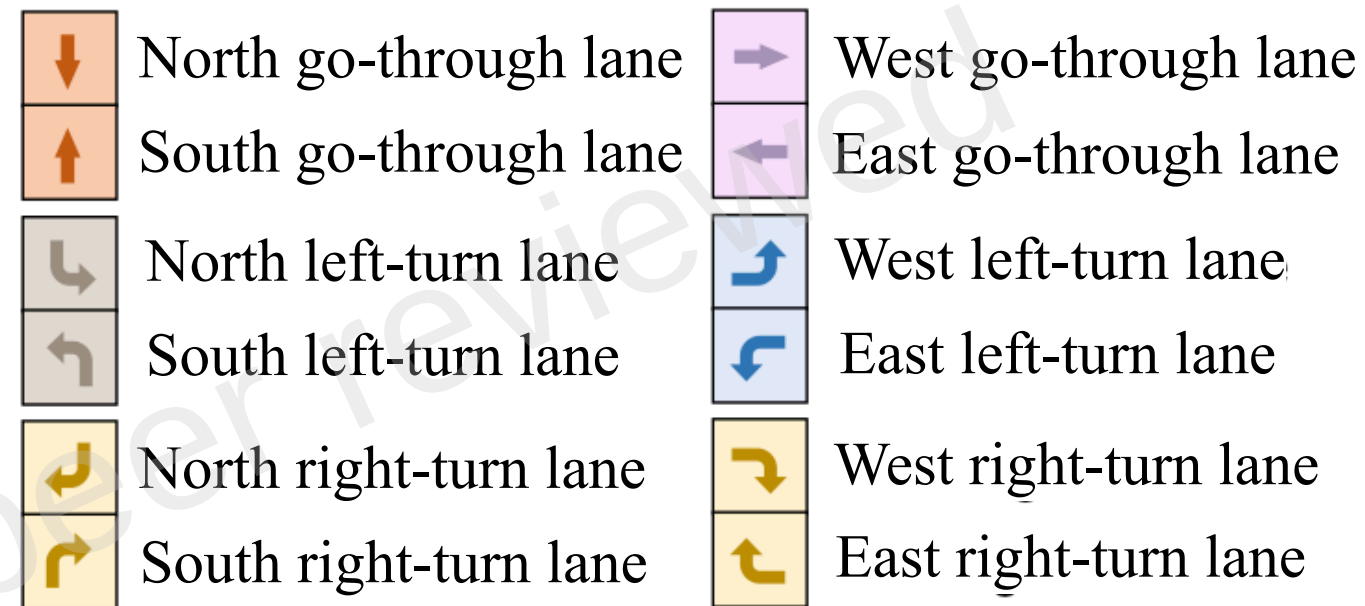




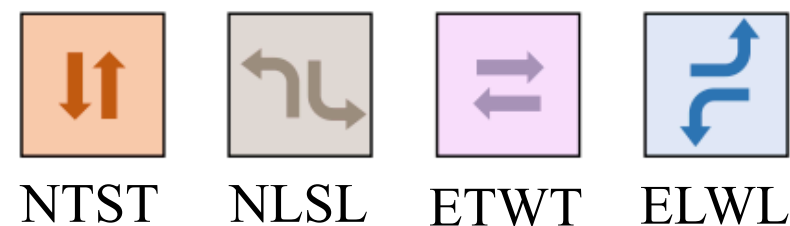




(a) Intersection



(b) North-South Lanes (c) East-West Lanes



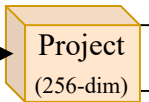
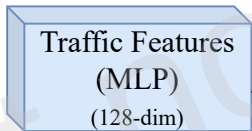
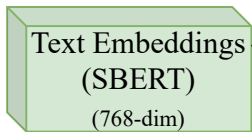
(d) Signal Phases

Input

Projection

Gate Computation

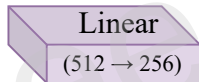
Output



text-proj
(Linear 768 $\rightarrow$ 256)



traffic-proj
(Linear 128 $\rightarrow$ 256)



concat

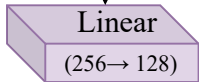
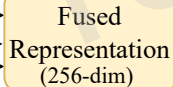


[Text; Traffic]
512-dim

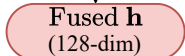


Gated g

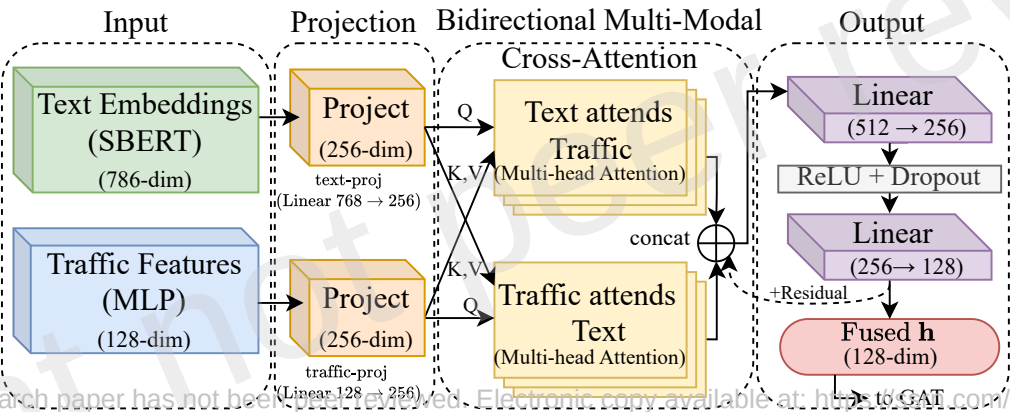
$$g \in [0, 1]^{256}$$



ReLU + Dropout



to GAT



Unstructured
Event Text



TRAFFIC
NETWORK
(4x4 Grid)

0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Attention
Aggregation

Input

Fused
Features

Multi-heads

$h=0$

$h=1$

...

$h=8$

Multi-heads

Multi-head
Attention

Fully
Connected

Output
Feature

Q-Network

